Abstract

Contents

Introduction

Massive black holes (MBHs) play a central role in the formation and evolution of galaxies. They are found at centers of nearby galaxies (Kormendy and Richstone 1995), and their masses are closely linked to the properties of the host galactic bulge as pointed out by Magorrian et al. (1998). MBHs are belived to be common in galaxy centres at all cosmic epochs; a piece of evidence for their existence, from low to high redshift, comes from the transient activity of galactic nuclei (AGN) over the hosts lifetime (Jahnke 2025).
Black holes (BHs) span a wide range of masses ($\sim 10 - 10^9\, M_\odot$) and various formation models have been proposed depending on the BH final mass. At the two extremes of the BH mass function, we find stellar BHs and supermassive black holes (SMBHs). Stellar evolution theory describes stellar BHs ($\lesssim 100\, M_\odot$) as the relic of massive stars, lighter then the trashold imposed by the pair-instability supernovae (Farmer et al. 2019); the properties of this population have been recently constrained through both the detection of gravitational waves (GWs) from merging binaries (Abbott et al. 2016) and the development of new microlensing analysis techniques (Bennett et al. 2002). On the opposite side ($\sim 10^6 - 10^9\, M_\odot$), instead, Pulsar timing arrays (PTA) as already probed the upper mass range of SMBHs population (Taylor 2021) and the most accepted model interpret them as the final stage of a secular accreation onto initially lighter black hole seeds.
Therefore, while substantial progresses have been achieved in detecting and characterizing both stellar BHS and SMBHs, the identification and study of intermediate-mass black holes, lying in the mass range $\sim 10^4 - 10^6\, M_\odot$, considered as possible SMBH seeds candidate (Devecchi and Volonteri 2009), still remain considerably more challenging, in particular at high redshift, due to the high spatial resolution required. A simple extrapolation of BH mass-host stellar bulge mass suggests that these elusive, compact objects should reside in the central region of dwarf galaxies (DWs) galaxies, with total luminosity below $5 \cdot 10^8\, L_\odot$, as confirmed by Reines et al. (2013).
Within the framework of galaxy hierarchical growth scenario, the creation of massive black hole binaries (MBHBs) is expected to occur, over cosmic time, as a natural consequence of mergers between galaxies. These systems are among the most powerful emitters of gravitational waves (GWs) in the universe, making them prime targets for detection with current and future GW observatories.
Although the merger rate of low mass galaxies has not been well determined yet, the interaction and merging of these structures should lead to the formation of IMBH-IMBH binaries. The upcoming Laser Interferometer Space Antenna (LISA), a joint ESA/NASA space mission, promises to demystify IMBHs, being sensitive to GW signals, from $\sim 10^{-4}$ Hz to $\sim 10^{-1}$ Hz, produced at late stage of the merger (Amaro-Seoane et al. 2017); their masses, within $10^4 - 10^7\, M_\odot$, and redshift will be estimated out to $z = 20$ with very high precision (Colpi et al. 2024).
Despite dwarf galaxies have been identified at incredibly high redshift (Kikuchihara et al. 2020), hence opening to the possibility of reciving GW signals from merging IMBH binaries if this

process takes place within the Hubble time, numerical simulations (Khan, Javed, et al. 2024) proved that in non-nucleated dwarfs the low stellar density in the central regions results in inefficient hardening and stalling of the binary. However, the vast majority of galaxies with stellar mass $10^8 - 10^{11}\, M_\odot$ harbor a nuclear star cluster (NSC), an extraordinarily dense and dynamically distinct stellar assembly, in their innermost regions; the existence of a such dense envirnoment inside dwarf galaxies, as reported by e.g. Nguyen et al. (2018), may drastically reduce the coalescence time of the IMBH pair.

This thesis work is aimed at studying the evolution of the main orbital parameters of the binary, with focus on semimajor axis a and eccentricity e, and analyzing how the variation of BHs masses, host galaxy properties and IC of the binary, i.e. (a_0, e_0) impacts , impact its evolution and coalescence time. In this spirit, the study combines a self-consistent hybrid model developed by Sesana (2010) with N-body simulations performed using the KETJU code (Mannerkoski et al. 2023), which will be described in more detail later.

The structure of this thesis is outlined as follows.

In chapter 1, we provide a brief overview of the main physical mechanisms responsible for the BH binary shrinking; this includes dynamical friction, stellar scattering processes and GWs emission,which dominates the late stages of the inspiral.

In chapter 2, we introduce the self-consistent hybrid model developed by Quinlan (1996) and Sesana (2010); it is thus applied to a sample of galaxies and exploring the parameters space in order to emphasise how different environmental and initial conditions lead to different binary evolution.

In chapter 3 we describe the main features of the codes used to generate N-body initial condition and N-body numerical simulations and compare the obtained binary evolutionary tracks with those of the previous chapter.

Finally, the main results of this work are summarized, and possible future improvements are reported in the concluding chapter.

1

Physical Mechanisms of Binary Hardening

This chapter aims to review the main physical processes involved in the MBH binary formation, from galaxy merging to the possible coalescence, presented in chronological order. It begins by addressing dynamical friction, effective down to the formation of a gravitationally bound pair, and by providing dynamical friction timescales for stellar envirnoments of astrophysical interest. Then, the main features of the gravitational slinghot will be discussed, as it rules the binary hardening via stellar ejection, along with the importance of loss-cone refilling in driving the system toward the inspiral phase.
The chapter ends with a derivation of the key equations describing the binary's time evolution once it enters the gravitational-wave-dominated phase, leading to its final coalescence.

1.1 Two-Body Relaxation and Dynamical Friction

In the "bottom-up" scenario, where low-mass structures form earlier and more massive ones assembly later through clustering, galaxy collisions represent a fundamental ingredient for MB-HBs creation. Being intrinsically inelastic, galaxy encounters, in particular those involving galaxies with comparable masses, give rise to mergers, leaving behind remnants hosting an MBH binary (see e.g. Baumgardt et al. (2006)).
During a galactic encounter, a fraction of the relative orbital energy of the two stellar systems is redistributed into the random motions of their constituent particles. This results in: heating of particles in the direction perpendicular to the initial relative velocity, promoting orbits deflection, and dynamical friction which slows down particles and drives the satellite galaxy most massive elements, e.g. MBH, towards the center of the primary galaxy.
Let us begin with the former effect; it naturally sets a timescale, the so called two-body relaxitation time t_{2b}, beyond which gravitational encounters alter stellar trajectories driven by the potential generated by $\rho(r)$, proxy of the real distribution of the galaxy stars. In line with that, systems whose age does not exceed the associated two-body relaxitation time, i.e. $t_{age} < t_{2b}$, will be referred to as collisionless, otherwise we talk about collisional systems and granularity will be important.
Chandrasekhar (1943) approach gives an estimate of t_{2b} considering cumulative orbital deflections under the Impulsive Approximation; in words:

1. All stellar encounters are indipendent of the others, i.e. multiple interactions are suppressed.

2. The gravitational influence of the remaining $N-2$ stars is negletted, resulting in hyperbolic two-body encounters only.

The first tool we need is the one-particle phase-space distribution function (DF) of the field stars, whichprovides the probability density of finding a particle at point $((\mathbf{x_f}, \mathbf{v_f})$ of the phase-space at time t. By definition, the stellar density is thus

$$\rho(\mathbf{x_f}, t) = \int_{\mathcal{R}^3} d^3\mathbf{v_f} f(\mathbf{x_f}, \mathbf{v_f}, t). \tag{1.1}$$

Under the simplifying assumption of a spatially homogeneous distribution and an isotropic velocity field, the degrees of freedom (d.o.f.) are reduced, and the distribution function can be expressed as

$$f(\mathbf{x_f}, \mathbf{v_f}) = n_f g(v_f); \tag{1.2}$$

hence, the DF has become the product of the position-indipendent number density and probability density function g in the velocity space
Two-body encounters between the test particle m_t and the field star m_f will induce change in some orbital property P. The cumulative effect of two-body encounters leads to a time variation of P, which is quantified by the diffusion coefficient $D(P)$; the Impulsive Approximation enables us ti write it as

$$D(P) \equiv 2\pi n_f \int_{\mathcal{R}^3} d^3\mathbf{v_f} \|\mathbf{v_t} - \mathbf{v_f}\| g(v_f) \int_0^\infty db \langle P \rangle b, \tag{1.3}$$

where $\langle P \rangle$ indicates the angle average and b the impact parameter.
A usefull indicator of changes in the stellar trajectories may be the orbital property $P = \Delta \mathbf{v_{t\perp}}$, that is the component of the test particle velocity variation perpendicular to the initial relative velocity. However, under the assumption of isotropy, the first-order diffusion coefficients in

the velocity components vanish (Binney and Tremaine 2011). Therefore, at higher order, the quantity $P = \|\Delta v_{t\perp}\|^2$ becomes the relevant orbital parameter for determining t_{2b} at leading order. Taking advantage of the Born aproximation of small angle scattering, holding for small deflection angles,

$$\|\Delta v_{t\perp}\| \simeq \frac{2Gm_f}{vb} \tag{1.4}$$

and plugging it into Equation (1.3) returns

$$D(\|\Delta v_{t\perp}\|^2) \simeq 8\pi G^2 m_f^2 n_f \int_{\mathcal{R}^3} d^3\mathbf{v_f} \frac{g(v_f)}{v} \int_0^\infty \frac{db}{b}. \tag{1.5}$$

Two logarithmic divergences arise from the integral over b: the so-called ultraviolet divergence at $b = 0$, due to the Born approximation, and the infrared divergence at $b = +\infty$ because of the used model, i.e. an infinite and homogeneus system. Neverthless, Adopting $b_{\pi/2}$ as lower limit and introducing a fiducial upper limit b_{max}, that we expect to exist due to the finite size of real stellar systems, it follows

$$D(\|\Delta \mathbf{v}_{t\perp}\|^2) \sim 8\pi G^2 n_f m_f^2 \int_{\mathbb{R}^3} \frac{g(v_f)\ln\Lambda}{v} \, d^3\mathbf{v}_f, \tag{1.6}$$

where $\ln(\Lambda) = \ln(b_{max}/b_{\pi/2})$, is the so called Coulomb logarithm. Although it depends on the velocity field, through the impact parameter $b_{\pi/2}$ corresponding to a deflection angle of $\pi/2$, it can be defined the velocity-weighted Coulomb logarithm $\ln(\bar{\lambda})$ so that

$$\int_{\mathbb{R}^3} \frac{g(v_f)\ln\Lambda}{\|\mathbf{v}_t - \mathbf{v}_f\|} \, d^3\mathbf{v}_f \equiv \Psi(v_t)\ln\bar{\Lambda}, \tag{1.7}$$

with

$$\Psi(v_t) = \int_{\mathcal{R}^3} \frac{g(v_f)}{\|\mathbf{v_t} - \mathbf{v_f}\|} d^3\mathbf{v_f}, \tag{1.8}$$

being the Rosenbluth potential. One of the greatest semplifications brung by isotropy consist of the possibility to express $\Psi(v_t)$ as

$$\Psi(v_t) = \int_{\mathbb{R}^3} \frac{g(v_f)}{\|\mathbf{v}_t - \mathbf{v}_f\|} \, d^3\mathbf{v}_f \equiv \frac{\Xi(v_t)}{v_t} + 4\pi \int_{v_t}^\infty g(v_f) v_f \, dv_f, \tag{1.9}$$

where

$$\Xi(v_t) = 4\pi \int_0^{v_t} g(v_f) v_f^2 \, dv_f, \qquad \Xi(\infty) = 1 \tag{1.10}$$

is a normalized function by virtue of the $g(v_f)$ function normalization.
In the high velocity limit, the monopole term of Equation (1.9) is the dominant to the leading order and

$$t_{2b} \equiv \frac{v_t^2(-\infty)}{D(\Delta E_{t\perp})} \sim \frac{v_t^3}{8\pi G^2 n_f m_f^2 ln\bar{\Lambda}}. \tag{1.11}$$

On one hand, as just seen, two-body encounters give rise to the heating of the test particle in the direction perpendicular to the initial relative velocity, being deflected; on the other hand the same particle is slowed down by Dynamical friction as result of long range interaction Baumgardt et al. 2006.

$$\frac{d\mathbf{v_t}}{dt} = D[\Delta \mathbf{v_t}] = -4\pi G^2 m_t m_f \ln\Lambda \int d^3\mathbf{v_f} f(\mathbf{v_f}) \frac{\mathbf{v_t} - \mathbf{v_f}}{|\mathbf{v_t} - \mathbf{v_f}|^3} \tag{1.12}$$

$$\frac{d\mathbf{v_t}}{dt} = -16\pi^2 G^2 m_t m_f \ln\Lambda \left[\int_0^{v_t} dv_f \, v_f^2 f(v_f)\right] \frac{\mathbf{v_t}}{v_t^3} \tag{1.13}$$

$$\frac{d\mathbf{v_t}}{dt} = -4\pi \ln\Lambda \left[erf(X) - \frac{2X}{\sqrt{\pi}} e^{-X^2}\right] \frac{G^2 m_t \rho(r)}{v_c^3} \mathbf{v_t} \tag{1.14}$$

where

$$erf(X) = \frac{2}{\sqrt{pi}} \int_0^X dt\, e^{-t^2} \tag{1.15}$$

$$\frac{d\tilde{L}}{dt} = \tilde{F} r \tag{1.16}$$

with

$$\tilde{L} = m_t \sqrt{G m_f r} \qquad \tilde{F} = -C(\Lambda, X) \frac{G^2 m_t^2 \rho(r)}{v_c^2} \tag{1.17}$$

with $C(\Lambda, X)$ implicitly defined by Equation (1.17) and $X \equiv v_c/\sigma\sqrt{2}$

$$r^{-5/2} \frac{dr}{dt} = -C(\Lambda, X(r)) \frac{G^2 m_t^2}{m_f^{3/2}} \rho(r) \tag{1.18}$$

$$\rho(r) = \begin{cases} \rho_{\rm inf} \left(\dfrac{r}{r_{\rm inf}}\right)^{-\gamma} & \text{if } r < r_{\rm inf} \\ \dfrac{\sigma^2}{2\pi G r^2} & \text{if } r > r_{\rm inf} \end{cases} \tag{1.19}$$

$$\rho(r) = \rho_0 \left(\frac{r}{r_0}\right)^{\gamma} \tag{1.20}$$

If $M_*(r < a_0) = 2M_2$ then

$$a_0 = (3-\gamma) \frac{GM}{\sigma^2} \left(\frac{q}{1+q}\right)^{\frac{1}{3-\gamma}} = r_{\rm inf} \left(\frac{q}{1+q}\right)^{\frac{1}{3-\gamma}} \tag{1.21}$$

$$\sigma^2(r) = \frac{v_c^2(r)}{2(\gamma - 1)} \quad \Rightarrow \quad X = \sqrt{\gamma - 1} \tag{1.22}$$

Integrating from r_0 to $r(t)$ leads to

$$r(t) = \begin{cases} \left[r_0^{\gamma-3/2} - C(\Lambda, X) \frac{\sqrt{G} m_t}{m_f^{3/2}} \left(\gamma - \frac{3}{2}\right) \rho_0 r_0^{\gamma} t\right]^{\frac{1}{\gamma - 3/2}} & \text{if } \gamma \neq \frac{3}{2} \\ r_0 \exp\left[-C(\Lambda, X) \frac{\sqrt{G} m_t}{m_f^{3/2}} \rho_0 r_0^{3/2} t\right] & \text{if } \gamma = \frac{3}{2} \end{cases} \tag{1.23}$$

If we define t_{fric} such that $r(t_{fric}) = a_0$ we get

$$t_{fric} = \begin{cases} \frac{m_f^{3/2} (r_{inf}^{\gamma-3/2} - a_0^{\gamma-3/2})}{C(\Lambda,X)\sqrt{G} m_t \left(\gamma - \frac{3}{2}\right) \rho_{inf} r_{inf}^{\gamma} (\gamma - 3/2)} & \text{if } \gamma \neq \frac{3}{2} \\ \frac{m_f^{3/2}}{C(\Lambda,X)\sqrt{G} m_t \rho_0 r_0^{3/2}} \ln\left(\frac{r_{inf}}{a_0}\right) & \text{if } \gamma = \frac{3}{2} \end{cases} \tag{1.24}$$

1.2 Gravitational Slingshot

One of the major results from the restricted three-body problem study consists of the slingshot effect Mikkola and Valtonen 1992. It plays an important role in shaping hyperbolic stellar orbits, unbound to a BH binary, and driving the binary shrinking.
Let m_t and m_f be the test and field masses; in the mass center frame S_{CM} the kinetic term related to $\|\mathbf{V_{CM}}\|$ vanishes and the total energy becomes

$$E = \frac{1}{2}\mu\|\mathbf{v}\|^2 - \frac{Gm_f m_t}{r}, \tag{1.25}$$

where $r \equiv \|\mathbf{x_t} - \mathbf{x_f}\| = +\infty$ and $\mathbf{v} \equiv \mathbf{v_t} - \mathbf{v_f}$ are particles relative separation and velocity respectively.
The energy conservation at two infinity configuarations, at time $t = \pm\infty$, leads to

$$\|\mathbf{v}(-\infty)\| = \|\mathbf{v}(+\infty)\|. \tag{1.26}$$

Eq. (1.26) implies that, in the inertial frame with the origin coincident with m_f position, m_t velocity vector $\mathbf{v}(t)$ has rotated and its variation $\Delta\mathbf{v} = \mathbf{v}(+\infty) - \mathbf{v}(-\infty)$ over the interaction can be decomposed as the sum of the components perpendicular and parallel to the initial relative velocity.
Therefore,

$$\mathbf{v}(+\infty) = \mathbf{v}(-\infty) + \Delta\mathbf{v}_{\parallel} + \Delta\mathbf{v}_{\perp} \tag{1.27}$$

and combining it with identity (1.26) follows:

$$\|\Delta\mathbf{v}_{\parallel}\|^2 + \|\Delta\mathbf{v}_{\perp}\|^2 + \langle\Delta\mathbf{v}_{\parallel}, \mathbf{v}(-\infty)\rangle = 0. \tag{1.28}$$

For Eq. (1.28) to hold, the scalar product must be necessarily negative.
These results, valid in a specific inertial frame, need to be transferred in an arbitrary inertial frame S_0, where both bodies may be seen moving. Here, the gravitational interaction causes both E_t and E_f to change but keeping $E = E_t + E_f = const$; specifically:

$$\Delta E_t = \frac{1}{2}m_t[\|\mathbf{v_t}(+\infty)\|^2 - \|\mathbf{v_t}(-\infty)\|^2] = -\Delta E_f. \tag{1.29}$$

Taking advantage of velocity composition law

$$\mathbf{v_t}(t) = \mathbf{v_{CM}} + (\mu/m_t)\mathbf{v}(t) \tag{1.30}$$

and the conservation of $\mathbf{v_{CM}}$, being the system isolated, E_t (1.29) takes the form

$$\Delta E_t = \mu\langle\Delta\mathbf{v}, \mathbf{v_{CM}}\rangle. \tag{1.31}$$

What derived so far can now be applied to the astrophysical system of interest.
Let us consider a BH binary-star gravitational encounter and be M_1, M_2 and m_* the primary, secondary BH and stellar masses respectively.
If BH binary is strongly asymmetric, that is $q \equiv M_2/M_1 \ll 1$, then

$$\mathbf{x_{CM}} \simeq \mathbf{x_f}, \qquad \mathbf{v_{CM}} \simeq \mathbf{v_f} \tag{1.32}$$

and M_2 will orbit approximately around M_1.
We focus on the interaction between the secondary BH and a star bound to M_1 and unbound to M_2. These objects will travel along nearly Keplerian orbits and have Keplerian velocity profile.
Hereafter, by analogy, we put $M_2 \equiv m_f$ and $m_* \equiv m_t$.
In the inertial frame centered at the primary BH, namely S_0, which behavies very similarly to S_{CM}, the interaction involving m_t and m_f is assumed to be short compared to the orbital period; this semplification makes negligible the acceleration of m_f-m_t mass center.
Finally, two different scenarios, giving rise to different feedback on binary evolution, will be studied.

1.2.1 Slingshot–Driven Binary Hardening

Working on the same model introduced at the end of Section 1.2, binary hardening results from gravitational slingshot when the stellar orbit is slightly bigger than secondary BH is.
Obeying to the Keplerian velocity profile, $\|\mathbf{v_f}(-\infty)\| > \|\mathbf{v_t}(-\infty)\|$. Additionally, as $m_f \gg m_t$, it holds the approximation $\mathbf{v_{CM}} \approx \mathbf{v_f}$; as a conequence of that, $\mathbf{v}(-\infty)$ is nearly antiparallel to $\mathbf{v_{CM}}$ and $\Delta\mathbf{v}_\parallel$ can be read as the component along $\mathbf{v_{CM}}$ rather than $\mathbf{v}(-\infty)$.
According to Eq. (1.28), we can conclude that

$$0 < \langle \Delta\mathbf{v}_\parallel, \mathbf{v_{CM}} \rangle = \langle \Delta\mathbf{v}, \mathbf{v_{CM}} \rangle = \Delta E_t/\mu. \tag{1.33}$$

Therefore, the star gains energy in S_0 and is pushed outward, at lareger separation from the primary BH; it is worth of noticing tath if $\Delta E_t > |E_t(-\infty)|$, stellar ejection occurs.
At same time, as dictated by energy conservation, the secondary BH loses energy, thus the orbit shriks and binary hardensCiotti 2021.

1.2.2 Slingshot–Driven Binary Softening

Binary softening insted takes place when the secondary BH orbit is outside the stellar one.
By repeating the same arguments, now $\mathbf{v}(-\infty)$ is nearly directed as $\mathbf{v_{CM}}$ and

$$E_t/\mu = \langle \Delta\mathbf{v}, \mathbf{v_{CM}} \rangle = \langle \Delta\mathbf{v}_\parallel, \mathbf{v_{CM}} \rangle < 0. \tag{1.34}$$

On one hand the star loses energy to the binary, it is pushed inward and may undergo tidal disruption or coalescence (stellar consumption) due to primary BH; on the other hand the binary gains energy and its separation increases (binary softening).

1.3 Stellar Scattering and Loss Cone Problem

Lightman and Shapiro 1977Let us consider a massive object M- it may be a BH binary- residing at center of a stellar distribution; far enough from it, stars are unbound, move with rms velocity $\langle v^2\rangle^{1/2}$ and obey to the Maxwellian distribution, i.e. velocity field is isotropic.
Combining M and the rms velocity, we define the influence radius, beyond which stars are mainly unbound, $r_a \sim 2GM/\langle v^2\rangle$. Stellar orbits are assumed to be driven by the gravity of M at $r < r_a$.
Moreover, close encounters between stars and M may lead to stellar consumption via several processes; they occur on a time scale of the order of the dynamical one, i.e. one orbital period, if r is smaller than the tidal radius r_t, where

$$t_d = P(E) = \frac{2\pi GM}{(2|E|)^{3/2}}. \tag{1.35}$$

A steady state solution in the range $r_t < r < r_a$ requires the rate of stars diffusing inward, crossing r_a, over t_{2b}, equal to the rate of stars consumed at r_t, behaving as a sink of this problem. If E and J denote respectively the star specific energy and angular momentum, it is worth of noticing that not only stars with low energy, i.e. $E \approx E_t \equiv -GM/r_t$ are removed but also high E low J might experience the same fate. At pericenter the radial velocity vanishes and

$$E = \frac{J^2}{2r^2} - \frac{GM}{r}; \tag{1.36}$$

hence a star can find itself inside r_t if

$$J \leq J_{min} \equiv \left[2\left(E + \frac{GM}{r_t}\right)\right]^{1/2} r_t. \tag{1.37}$$

Inequality (1.37) defines the so called "Loss Cone".

1.3.1 Bound stars consumption

One-Dimensional Analysis

As stellar orbit shape and size around M are set by E and J, the one particle DF can be written as $f = f(E, J)$ according to Jean's theorem. Therefore, the nature of the problem is 2-dmensional; neverthless, by negletting loss cone effects, an approximated one-dimensional solution can be obtained under the assumption of quasi-circular orbits where $J = J_{max}(E) = GM/\sqrt{2|E|}$.
In this scenario the boundary conditions for bound stars at $r_t < r < r_a$ read $E_t < E < E_a$ and the DF is $f = f(E)$. The steady stae self-similar solution obtained by Peebles relies on the power law DF $f(E) = k|E|^p$ and the corresponding maximum flux of stars in the E space is $F_{max} \sim N(< r)/t_{2b}(r)$.
By definition

$$N(< r) = \int f(E) d^3\mathbf{v} \sim |E|^{p-3/2} \tag{1.38}$$

and $t_{2b} \sim v^3/n \sim |E|^{-p}$; they combine into

$$F_{max}(E) \sim |E|^{2p-3/2}. \tag{1.39}$$

A more accurate analysis yields

$$F(E) = -F_{max}(E)\beta(p), \tag{1.40}$$

where the the exact solution, that is the value of p, can be derived by imposing $F(E) = const$ from the stationarity condition.
Bahcall and Wolf (1976) found the solution $p = 1/4$.

Two-dimensional Analysis: Loss Cone effects

Let us focus on loss cone effects; the DF is now in general a function of both E and J.
By assuming stellar encounters to be dominated by distant and weak gravitational interactions, J diffusion problem can be treated with RW theory and it can be defined the angular momentum step size

$$j_2 \sim J_{max}(t_d/t_r)^{1/2}, \tag{1.41}$$

it represents the dispersion in J suffered over t_d.
The dimensionless quantity

$$q(E) \equiv \frac{j_2^2}{J_{min}^2} \tag{1.42}$$

enables us to distinguish two extreme scenarios in terms of stellar consumption rate:

- $q \ll 1$ (Empty Loss Cone): $j_2 \ll J_{min}$ implies a too small RW step size for diffusion to be effective and only stars with J close to J_{min} can enter the loss cone.
  Since $t_{2b} \gg t_d$ the loss cone remeins nearly empty and the capture rate is limited by diffusion.

- $q \gg 1$ (Full Loss Cone): $j_2 \gg J_{min}$ implies a large RW step size compared to loss cone size and many stars in the range $J_{min} < J < J_{min} + j_2$ can be captured.
  In this regime equation (1.41) predicts many weak stellar encounters per orbital period and thus an almost full loss cone as the star consumption, takes place over t_d,is slower than stellar capture rate; stellar consumption rate is limited by orbital period.

The stellar consumption rate per unit energy is

$$-F_E(E) \sim \begin{cases} \frac{F_{max}(E)|E|^{-1}}{ln\left(\frac{GM}{|E|r_t}\right)}, & q \ll 1 \\ q^{-1}F_{max}(E)|E|^{-1}, & q \gg 1 \end{cases} \tag{1.43}$$

and is strongly peaked at E_{crit}- or equivalently at r_t- at which $q(E_{crit}) \approx 1$.

1.3.2 Unbound stars consumption

Previously it has been pointed out that the majority of the stellar capture involves stars at E_{cri} (or r_{crit}), this indicates that if the critical radius exceeds the influence radius, within which stars are mainly bound, this process affects primarly Unbound stars.
In this region the stellar flux becomes

$$F_{max} \sim \frac{N(<r)}{t_{2b}(r)} \sim \left(\frac{r}{r_a}\right)^3 F_{max}(E_a) \tag{1.44}$$

1.3.3 Loss cone theory role in Hybrid Model

Devoted to determine evolutionary tracks for BH binaries evolving via three-body scattering with background stars and GWs emission Quinlan 1996 Sesana 2010, one of the key points of the adopted model turns out to be the nature of stars involved in close encounters with the binary. At beginning, when the pair forms, all stars intersecting its orbit, i.e. populating the

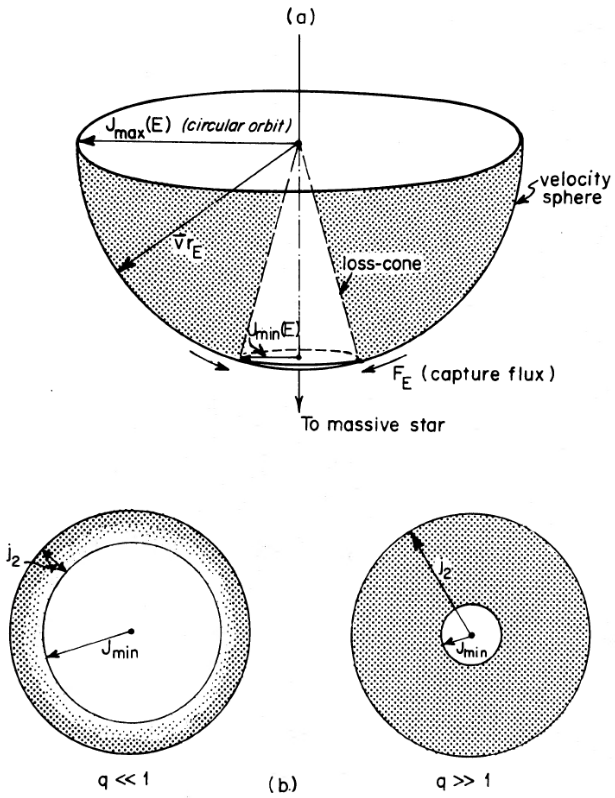


Figure 1.1: Fig. 1(a).—Velocity distribution of stars with fixed energy E;J varies accordingly to the direction of $\mathbf{v}$ emphasising the two-dimensional nature of the problem. The vertical axis points towards the compact object of mass M. Fig. 1(6).—The velocity sphere viewed from below. On the left the empty loss cone scenario; stellar density is nearly indipendent of J everywhere but $J \gtrsim J_{min}$ where is quickly declines. On the right the full loss cone regime; consumed stars are immediately replenished by stars from a region much more extended- and uniform density- than the loss cone itself (the "pinhole" limit).

"Loss cone" are expelled via gravitational slingshot effect mainly; this mechanism results in the erosion of the stellar core and drives efficiently the binary down to a separation of the order of the hardening radius. Thereafter, for the shrinking to proceed, loss cone orbits must be refilled. Without addressing the problem concerning how this process takes place, it is relevant to point out that, once depleted all the stars inside a_h, stars candidate to replenish the loss cone come from outside this region.

Therefore, for nearly equal mass BHs, having $a_h \sim r_{inf}$, or shallow stellar cores from the transistion soft/hard on, the scattering with unbound stars becomes dominant until the binary develops efficient GWs emission phase.

Once established whether the vast majority of stars exchanging energy and angular momentum with the binary are bound or unbound to such a pair, the Hybrid model should take into account loss cone status through some parameters. This information enters the model by means of the normalizing density and velocity dispersion of H_u and K_u; when they are set equal to their value at r_{inf}, the model is assuming that the loss cone is always full at $r > r_{inf}$ (pinhole regime) and empty at $r < r_{inf}$ (diffusive regime).

Testing the model by changing the normalizing quantities serves to investigate how the loss cone refilling affects the binary evolution.

1.4 Gravitational Radiation from Black Hole Binaries

1.4.1 Linearized Gravitational Field Equation

The starting point of the theoretical derivation of GWs properties, such as the energy and angular momentum they transport, from BH binary and the associated back-reaction on the astrophysical source, is the linearization of field equations of Generl Relativity (GR)

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{1.45}$$

These constitute ten coupled, non-linear, second-order partial differential equations for the metric tensor; neverthless, they are not sufficient to uniquely set the ten indipendent components of $g_{\mu\nu}$ due to the four Bianchi identities, equivalent to the same number of differential constraints, resulting in six independent equations only. These four degrees of freedom are intimately connected to gauge symmetries, reflecting the arbitrariness of the four coordinates x^μ used to represent and solve Equation (1.45).
To show how gravitational radiation emerges from Equation (1.45), we need to expand it around the Minkowski metric $\eta_{\mu\nu} = diag(-1,+1,+1,+1)$; therefore we assume

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + h_{\mu\nu}(x), \tag{1.46}$$

with $|h_{\mu\nu}| \ll 1$ being a symmetric tensor representing a small perturbation of the flat spacetime. The left-hand side of Equation (1.45) encodes the geometry and curvature of spacetime and contains all terms that are functions of the metric; in terms of the affine connection

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) \tag{1.47}$$

and the Riemann tensor

$$R_{\lambda\mu}{}^{\rho}{}_{\nu} = \partial_\lambda \Gamma^\rho_{\mu\nu} - \partial_\mu \Gamma^\rho_{\lambda\nu} + \Gamma^\rho_{\lambda\sigma}\Gamma^\sigma_{\mu\nu} - \Gamma^\rho_{\mu\sigma}\Gamma^\sigma_{\lambda\nu}, \tag{1.48}$$

they can be expressed indeed as

$$R_{\mu\nu} = R_{\sigma\mu}{}^{\sigma}{}_{\nu} \tag{1.49}$$

and

$$R = g^{\alpha\beta} R_{\alpha\beta} = R^\sigma_\sigma. \tag{1.50}$$

On the right-hand side, instead, appears the Energy-Momentum tensor; it can be read as the source term of the gravitational field equations and turns out to be

$$T^{\mu\nu} = (\rho + p/c^2) u^\mu u^\nu + p g^{\mu\nu} \tag{1.51}$$

for a perfect fluid of density ρ, pressure p and four-velocity v^μ.
It straightforward to prove that, to the first order in the metric perturbation, they become:

$$R_{\mu\nu} = -\frac{1}{2}(\Box h_{\mu\nu} + \partial_\mu\partial_\nu h - \partial_\lambda\partial_\mu h^\lambda_\nu - \partial_\nu\partial_\lambda h^\lambda_\mu) \tag{1.52}$$

and

$$R = \partial_\mu\partial_\nu h^{\mu\nu} - \Box h, \tag{1.53}$$

where h denotes $h_{\mu\nu}$ trace.
Defining the so-called "trace reverse" of $h_{\mu\nu}$

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu} h \tag{1.54}$$

and plugging the terms of Equations (1.52) and (1.53) into Equation (1.45), the field equations take the form

$$\Box\bar{h}_{\mu\nu} + \eta_{\mu\nu}\partial_\alpha\partial_\beta\bar{h}^{\alpha\beta} - \partial_\mu\partial_\lambda\bar{h}^\lambda_\nu - \partial_\nu\partial_\lambda\bar{h}^\lambda_\mu = -\frac{16\pi G}{c^4}T_{\mu\nu}. \tag{1.55}$$

Taking advantage of the gauge freedom of Equation (1.45), it can be performed an infintesimal coordinate transformation of the kind

$$x^\mu \rightarrow x'^\mu = x^\mu + \xi^\mu(x); \tag{1.56}$$

as tensors obey to precise law of transformation, this yields

$$\bar{h}'_{\mu\nu} = \frac{\partial x^\alpha}{\partial x'^\mu}\frac{\partial x^\beta}{\partial x'^\nu}\bar{h}_{\alpha\beta} \approx \bar{h}_{\mu\nu} - \partial_\mu\xi_\nu - \partial_\nu\xi_\mu + \eta_{\mu\nu}\partial_\sigma\xi^\sigma. \tag{1.57}$$

Yet, if the four functions ξ^μ are chosen so as to satisfy the so-called De Donder gauge condition, i.e. $\Box\xi^\mu = \partial_\sigma\bar{h}^{\sigma\mu}$, then $\partial_\sigma\bar{h}'^{\sigma\mu} = 0$ and, dropping the prime, Equation (1.55) semplifies to

$$\begin{cases} \Box\bar{h}_{\mu\nu} = -\dfrac{16\pi G}{c^4}T_{\mu\nu} \\ \partial_\mu\bar{h}^{\mu\nu} = 0. \end{cases} \tag{1.58}$$

We are now ready to address the problem of determining the general solution of the linearized Einstein's equations coupled with the De Donder gauge condition.

1.4.2 Generation and Propagation of Gravitational Waves

To study the formation of GWs we initially focus on the solution of the problem defined in Equation (1.58) inside the source where $T_{\mu\nu} \neq 0$.
Linear partial differential equations solution can be obtained by the method of Green's function; by definition, the Green's function G is the solution of the associated equation in case of a pont-like source, i.e.

$$\Box_x G(x - x') = \delta^{(4)}(x - x'), \tag{1.59}$$

with $\delta^{(4)}(x - x')$ being the four-dimensional Dirac function.
It is relevant to note that, apart from the B.C. of the problem, G depends only on the linear operator; therefore, from the radiation problem in electromagnetism, we already know that the proper solution is the retarded Green's function

$$G(x - x') = -\frac{1}{4\pi|\mathbf{x} - \mathbf{x}'|}\delta^{(4)}(ct_{ret} - ct'), \tag{1.60}$$

where

$$t_{ret} = t - \frac{|\mathbf{x} - \mathbf{x}'|}{c} \tag{1.61}$$

is the time in past at which the radiation was emitted at $\mathbf{x}'$ in order to be observed at $\mathbf{x}$ at time t. Once the Green's function is found, the general solution will be the convolution between it and source term:

$$\bar{h}_{\mu\nu}(t, \mathbf{x}) = \frac{4G}{c^4}\int d^3x' \frac{T_{\mu\nu}(t_{ret}, \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|}. \tag{1.62}$$

Prior to examining the formula reported in Equation (1.62) and assessing which additional simplifications can be implemented and when they apply, it is appropriate to analyze how GWs propagate in vacuo.

Outside the GWs source, $T_{\mu\nu}$ vanishes and, owing to the linearity of the d'Alembert operator, the solution can be expressed as superposition of plane waves without loss of generality, that is

$$\bar{h}_{\mu\nu}(x) = \int d^3k(A_{\mu\nu}(\mathbf{k})e^{ik_\rho x^\rho} + c.c.) \tag{1.63}$$

However, not all plane waves satisfy the linearized field equation in vacuo along with the gauge condition but linearity enables us to study the properties of a single plane wave separately; consequently, inserting the Ansatz $\bar{h}_{\mu\nu} \sim A_{\mu\nu}e^{ik_\rho x^\rho}$ yields the following two constraints:

1. ($k_\rho k^\rho = 0$): the four-vector $k^\mu = (\omega/c, \mathbf{k})$ is a light-like vector (GWs travel at light speed c in vacuo).
2. ($k_\mu A^{\mu\nu} = 0$): it implies $A^{0\nu} = A^{3\nu}$ if we deal with GWs propagating along the x^3 direction, hence $k^\mu = (k, 0, 0, k)$.

At this stage, the amplitude tensor $A^{\mu\nu}$ possesses six indipendet components — ten, due to symmetry, reduced by the four constraints imposed by the gauge conditions — and can be written as

$$A^{\mu\nu} = \begin{pmatrix} A^{00} & A^{10} & A^{20} & A^{00} \\ A^{10} & A^{11} & A^{12} & A^{10} \\ A^{20} & A^{12} & A^{22} & A^{20} \\ A^{00} & A^{10} & A^{20} & A^{00} \end{pmatrix}. \tag{1.64}$$

Out of the six polarizations shown in Equation (1.64), four are unphysical because they vanish when the residual gauge freedom associated with four arbitrary harmonic functions ξ^μ is exploited. One is free to adopt functions proportional to the plane wave solutions of the vacuo problem, i.e. $\xi^\mu = \epsilon^\mu e^{ik_\rho x^\rho}$ are harmonic, and to set their amplitudes ϵ^μ such that

$$A^{\mu\nu}_{TT} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_x & 0 \\ 0 & h_x & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{1.65}$$

The coordinate system where $A^{\mu\nu}$ is in the form of Equation (1.65), with two indipendent polarizations only, and defined by

$$\partial_\mu h^{\mu\nu} = 0, \quad h^{0\mu} = 0, \quad h^i_i = 0, \tag{1.66}$$

is referred to as "TT gauge"[1].

Having analyzed the propagation of GWs in vacuo and the properties of the TT gauge, we are now in the position to focus again on the source solution reported in Equation (1.62). One of the greatest simplifications to this integral consists of the so-called "Compact-source approximation"; it applies outside the source whenever its size is negligible with respect to the source-detector separation. In this regime, we can retain the leading term of the multipole expansion only, thus

$$h^{TT}_{ij}(t, \mathbf{x}) = \frac{4G}{c^4 r}\int d^3x' T_{ij}(t_{ret}, \mathbf{x}'). \tag{1.67}$$

Furthermore, far from the source, the conservation of the Energy-Momentum tensor, namely $\nabla_\mu T^{\mu\nu} \approx \partial_\mu T^{\mu\nu} = 0$, yields

$$h^{TT}_{ij}(t, \boldsymbol{x}) = \frac{2G}{rc^4}\ddot{Q}^{TT}_{ij}(t - r/c) \tag{1.68}$$

[1] In the TT gauge, according to the definition provided, $\bar{h}^{TT}_{ij} = h^{TT}_{ij}$.

at lowest order in the low-velocity expansion.
The term Q_{ij}^{TT} introduced in Equation (1.68)[2] is the representation of the quadrupole moment in the class of coordinate systems satisfying the TT gauge conditions; defining the quadrupole moment of $T^{00}/c^2 = \rho$ as

$$M^{ij} \equiv \frac{1}{c^2} \int d^3x T^{00}(t, \boldsymbol{x}) x^i x^j, \tag{1.69}$$

it is thus possible to write

$$Q^{ij} \equiv M^{ij} - \frac{1}{3}\delta^{ij} M_{kk} = \int d^3x \rho(t, \mathbf{x}) \left(x^i x^j - \frac{1}{3} r^2 \delta^{ij} \right). \tag{1.70}$$

Equation (1.67) shows that, unlike electromagnetic radiation, at lowest order the gravitational radiation arises from changes in the quadrupole moment; other important observables such as the energy and angular momentum carried by GWs, needed to estimate the back-reaction, will be derived from the quantities defined in Equation (1.70) indeed.
The energy and angular momentum conservation laws impose that the rates a which these quantities are carried away by GWs at time t and at large separation r from the source coincide with the corresponding rates of orbital energy —if the emitting system consists of a BH binary— and angular momentum, negletting the spin contribute, loss evaluated at the appropriate retarded time; in formulae:

$$P_{quad} = \frac{dE_{gw}}{dt} = \frac{G}{5c^5} \langle \dddot{Q}_{ij} \dddot{Q}_{ij} \rangle = -\frac{dE_{source}}{dt}, \tag{1.71}$$

or equivalently in terms of M_{ij}

$$P_{quad} = \frac{G}{5c^5} \left\langle \dddot{M}_{ij} \dddot{M}_{ij} - \frac{1}{3} (\dddot{M}_{kk})^2 \right\rangle, \tag{1.72}$$

and

$$\frac{dL^i_{gw}}{dt} = \frac{2G}{5c^5} \epsilon^{ijk} \langle \ddot{Q}_{jl} \dddot{Q}_{kl} \rangle = -\frac{dL^i_{source}}{dt}. \tag{1.73}$$

Equations (1.71) and (1.73) will be of fundamental importance for estimating the back-reaction induced by GWs emission on the black hole binary.

1.4.3 Gravitational Waves from Black Hole Binaries and Back-Reaction

We now focus on the gravitational radiation emitted by two masses m_1 and m_2 moving along elliptic Keplerian orbit. Let $m = m_1 + m_2$ and $\mu = m_1 \cdot m_2/m$ be the total mass and the reduced mass respectively.
Regardless of whether the orbit is bound or unbound, the motion of the two bodies takes place in a plane due to the conservation of the angular momentum $\mathbf{L}$; therefore, in this plane, the x-y plane, perpendicular to $\mathbf{L}$, it is convenient to introduce the polar coordinates (r, ψ) with origin at center-of-mass.
In this frame, as before denoted by S_{CM},

$$L_z = \mu r^2 \dot{\psi} \tag{1.74}$$

coincides with the modulus L of the costant vector while, as a consequence of the Köenig theorem,

$$E = \frac{1}{2}\mu \dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{Gm\mu}{r} \tag{1.75}$$

[2]The components with at least one time index are excluded from our analysis because they are constant and therefore do not contribute to GWs formation.

exhibits only the term of kinetic energy corresponding to the relative motion of the system, as in S_{CM} holds $K_{CM} = 0$ by definition.
The integration of Equation (1.75) requires, as in (Capozziello and De Laurentis 2008), the introduction of the auxiliary variable $w(\psi) = 1/r(\psi)$; plugging it into the energy conservation identity, Equation (1.75) becomes

$$w'^2 + w^2 - \frac{2Gm_1m_2\mu}{L^2}w = \frac{2\mu E}{L^2} \tag{1.76}$$

where $w' \equiv dw/d\psi$. Differentiating both sides of the last equation with respect to ψ leads to

$$w'\left(w'' + w - \frac{Gm_1m_2\mu}{L^2}\right) = 0. \tag{1.77}$$

Apart from the solution $w = const$, corresponding to a circular binary, the problem admits harmonic–oscillator–like and, upon solving for $r(\psi)$, it is found to be given by

$$r(\psi) = \frac{R}{1 + e\cos\psi}, \tag{1.78}$$

with $R = L^2/(Gm\mu^2)$ being semi-latus rectum — the polar axis is aligned to the major axis $2a$, so that R corresponds to the distance between the origin and the point of the conic at $\psi = \pi/2$— and is related to the eccentricity e by

$$e^2 = 1 - \frac{R}{a} = 1 + \frac{2EL^2}{G^2m^2\mu^3}. \tag{1.79}$$

It is worth of noticing that we are interested in the GWs emission process by a BH binary, which is by definition a gravitationally bound system; therefore, from this point onward it will be assumed $E < 0$ and thus, by virtue of Equation (1.79), $0 \leq e < 1$. Since the two semiaxes are given by

$$a = \frac{R}{1 - e^2} \qquad b = \frac{R}{\sqrt{1 - e^2}}, \tag{1.80}$$

inserting Equation (1.79) into the first identity of Equation (1.80) gives the semi-major axis in terms of E; explicitly:

$$a = -\frac{Gm\mu}{2E} \tag{1.81}$$

It follows that fixing the energy means setting the semi-major axis.
The functions $r(t)$ and $\psi(t)$ can be obtained by direct integration of Equations (1.74) and (1.75); they are usually given in the parametric form

$$r(t) = a[1 - e\cos u(t)] \qquad \psi(t) = \frac{\cos u(t) - e}{1 - e\cos u(t)} \tag{1.82}$$

in terms of the "eccentric anomaly" u defined in the famous Kepler equation

$$u(t) - e\sin u(t) = \left(\frac{Gm}{a^3}\right)^{1/2} t \equiv \omega_0 t. \tag{1.83}$$

The transformation from from polar cordinates (r, ψ) to Cartesian ones (x, y) is illistrated below

$$\begin{cases} x = r\cos\psi = a\left[\cos u(t) - e\right] \\ y = r\sin\psi = b\sin u(t). \end{cases} \tag{1.84}$$

We now possess all the necessary ingredients to proceed with the computation of the rates of energy and angular momentum carried by GWs, which will play a fundamental role in assessing the back-reaction on the binary system.
In the reference system S_{CM} holds $\rho(t, \mathbf{x}) = \mu\delta(x - r\cos\psi)\delta(y - r\sin\psi)\delta(z)$ and, according to Equation (1.69), it is immediate to get

$$M_{ij} = \mu r^2(t) \begin{bmatrix} \cos^2(\psi) & \sin(\psi)\cos(\psi) \\ \sin(\psi)\cos(\psi) & \sin^2(\psi) \end{bmatrix}. \tag{1.85}$$

The time derivatives are easier computed when r, which is a function of time itself, is expressed as a function of ψ; this is accomplished by replacing this variable with the general solution of the two-body problem provided by Equation (1.78). A simple computation returns the third time derivatives needed to derive the power radiated in the quadrupole approximation, i.e.

$$\dddot{M}_{11} = \beta(1 + e\cos\psi)^2 \left[2\sin 2\psi + 3e\sin\psi\cos^2\psi\right], \tag{1.86}$$

$$\dddot{M}_{22} = \beta(1 + e\cos\psi)^2 \left[-2\sin 2\psi - e\sin\psi(1 + 3\cos^2\psi)\right], \tag{1.87}$$

$$\dddot{M}_{12} = \beta(1 + e\cos\psi)^2 \left[-2\cos 2\psi + e\cos\psi(1 - 3\cos^2\psi)\right], \tag{1.88}$$

where the factor β is defined as

$$\beta^2 \equiv \frac{4G^3\mu^2 m^3}{a^5(1 - e^2)^5}. \tag{1.89}$$

By means of Equation (1.71) we get in conclusion

$$P(\psi) = \frac{8G^4}{15c^5}\frac{\mu^2 m^3}{a^5(1 - e^2)^5}(1 + e\cos(\psi))^4[12(1 + e\cos(\psi))^2 + e^2\sin^2(\psi)]. \tag{1.90}$$

However, we recall that this istantaneous quantity is not constant over one orbital period for elliptic Keplerian orbits as in general M_{ij} will suffer from different rate of variation from pericenter to apocenter; therefore, taking the temporal average over one orbital period, $T \equiv 2\pi/\omega_0$ returns

$$P \equiv \frac{1}{T}\int_0^T dt P(\psi) = \frac{\omega_0}{2\pi}\int_0^{2\pi} d\psi \frac{P(\psi)}{\dot{\psi}} = \frac{32G^4\mu^2 m^3}{5c^5 a^5}F(e), \tag{1.91}$$

with

$$F(e) = \frac{1}{(1 - e^2)^{7/2}}\left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right) \tag{1.92}$$

being a function strongly depending on the eccentricity, which reduces to unity in case of circular binary.
Throughout the inspiral phase, the binary radiates not only energy but angular momentum as well, carried by GWS, and undergoes secular changes in both semimajor axis and ellipticity until coalescence eventually occurs. Therefore, another key ingredient in evaluating the back-reaction on the binary evolution is the determination of the angular momentum loss rate. It is quantified in Equation (1.73) which becomes

$$\frac{dL^i_{source}}{dt} = -\frac{2G}{5c^5}\epsilon^{ikl}\left\langle \ddot{M}_{ka}\dddot{M}_{la}\right\rangle \tag{1.93}$$

as the the terms of the form $\epsilon^{ijk}\delta_{jl}\dddot{Q}_{kl}$ vanish, since they correspond to the contraction of an antisymmetric tensor with a symmetric one.
Taking into account that, in S_{CM}, all components M_{i3} are identically zero and that $L = L_z =$

L^3_{source}, one can rewrite Equation (1.93) in a more convenient form consistent with the previous arguments:

$$\frac{dL}{dt} = \frac{4G}{5c^5}\langle \ddot{M}_{12}(\dddot{M}_{11} - \dddot{M}_{22})\rangle. \tag{1.94}$$

The last identity provides dL/dt in terms of the second and third derivatives of the quadrupole moment of ρ; although the second derivative $\ddot{M}_{12}$ has not been computed explicitly yet, it follows straightforwardly from the same procedure used to derive $\dddot{M}_{ij}$, hence

$$\ddot{M}_{12} = \frac{G\mu m}{a(1-e^2)}\sin\psi\left[-4(1+e\cos\psi)^2\cos\psi + 2e\left(3\cos^2\psi - 1 + 2e\cos^3\psi\right)\right]. \tag{1.95}$$

Yet, adopting the expressions for reported in Equations (1.86), (1.87) and (1.95) respectively, we get the angular momentum loss, corresponding to minus the angular momentum carried by the gravitational radiation in the quadrupole approximation; then, the temporal average over a single orbital period T, transformed into an integral over the angular position ψ, returns

$$\frac{dL}{dt} = -\frac{32}{5}\frac{G^{7/2}\mu^2 m^{5/2}}{c^5 a^{7/2}}\frac{1}{(1-e^2)^2}\left(1+\frac{7}{8}e^2\right) \tag{1.96}$$

Lastly, the simi-major axis and eccentricity evolution of the BH binary results by imposing the conservation of the energy and angular momentum of the system composed of the binary and GWs; differentiating Equations (1.79) and (1.81) with respect to the time yields

$$\frac{da}{dt} = \frac{Gm\mu}{2E^2(a)}(-P) \tag{1.97}$$

and

$$2e\frac{de}{dt} = \frac{2}{G^2m^2\mu^3}\left[-PL^2(a,e) + 2E(a)L(a,e)\frac{dL}{dt}\right]. \tag{1.98}$$

By rearranging all the terms and performing the appropriate substitutions, Peters and Mathews (1963) gave

$$\frac{da}{dt} = -\frac{64}{5}\frac{G^3\mu m^2}{c^5a^3}\frac{1}{(1-e^2)^{7/2}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \tag{1.99}$$

along with

$$\frac{de}{dt} = -\frac{304}{15}\frac{G^3\mu m^2}{c^5a^4}\frac{e}{(1-e^2)^{5/2}}\left(1+\frac{121}{304}e^2\right) \tag{1.100}$$

that has been employed by Quinlan (1996) and Sesana (2010) in order to reconstruct evolutionary tracks of BH binaries through a "Hybrid model", which will be described later.
Equations (1.99) and (1.100) prove that GWs emission is efficient, more precisely at little binary separation due to a^{-3} and a^{-4} dependence respectively, not only in shrinking the BHs pair, by extracting orbital energy, but in promoting the orbit circularization as well, thus making this mechanism dominant over stellar scattering at late stage of binary inspiral.
Furthermore, another key piece of information that can be derived from the theory is the expected coalescence time τ of the binary in the case of gravitational-wave emission alone. This should reproduce the characteristic hardening timescale of the final stage, when orbital shrinking driven by gravitational radiation takes over stellar scattering. This is obtained by combining Equations (1.99) and (1.100); this mathematical operation leads to a differential equation for $a = a(e)$, which reads

$$\frac{da}{de} = \frac{12}{19}\frac{1+\frac{73}{24}e^2+\frac{37}{96}e^4}{e(1-e^2)\left(1+\frac{121}{304}e^2\right)}. \tag{1.101}$$

The last equation admits analytic solution of the kind

$$a(e) = a_0 \frac{g(e)}{g(e_0)} \tag{1.102}$$

where

$$g(e) = \frac{e^{12/19}}{1-e^2}\left(1+\frac{121}{304}e^2\right)^{870/2299} \tag{1.103}$$

and (a_0, e_0) denotes the initial point of the of binary evolutionary track.
Last but not least, given the BH binary initial condition (a_0, e_0), the coalescence time is found by integration over the eccentricity and by noticing that, if we set $(a(0), e(0)) = (a_0, e_0)$, $e(\tau) = 0$ due to orbit circularization:

$$\tau(a_0, e_0) = \int_0^{\tau(a_0,e_0)} dt = \int_{e_0}^{0} de \frac{dt}{de} = \int_{e_0}^{0} de \frac{a^4(e)(1-e^2)^{5/2}}{e\left(1+\frac{121}{304}e^2\right)}. \tag{1.104}$$

By substituting Equation (1.102) into Equation (1.104) one gets the final expression

$$\tau_{\rm coal}(a_0, e_0) = \frac{12}{19}\frac{c^5}{G^3}\frac{a_0^4}{\mu m^2}\int_0^{e_0} \frac{e^{29/19}\left(1+\frac{121}{304}e^2\right)^{1181/2299}}{(1-e^2)^{3/2}}\,\mathrm{d}e. \tag{1.105}$$

2

A Hybrid Approach to Binary Evolution

2.1 Quinlan Formalism

2.1.1 Introduction

In this work, binary evolution embedded in a fixed stellar background problem is addressed in a hybrid fashion providing a self consistent analytical framework where dimensioless rates are directely estimated from scattering experiments.
Scattering experiments are carried out for the restricted three-body problem and proceed until either the binary coalesce or stellar ejection has become so important that the host galaxy can no longer be considered fixed.
Three dimensionless quantities (H, K, J) are measured from numerical experiments and defined as:

$$H = \frac{\sigma}{G\rho}\frac{d}{dt}\left(\frac{1}{a}\right), \tag{2.1}$$

$$K = \frac{de}{d\ln(1/a)} \tag{2.2}$$

and

$$J = \frac{1}{M_{12}}\frac{dM_{ej}}{d\ln(1/a)}. \tag{2.3}$$

Equations (2.1), (2.2) and (2.3) introduce the so called dimensionless hardening rate H, eccentricity growth rate K and mass loss rate. Once (H, J, K) are obtained from scattering experiments, the system of coupled differential equations can be solved for the binary semi-major axis, eccentricity and ejected mass time derivatives and numerical methods will eventually return their evolution within the fixed galaxy model of interest.

2.1.2 Quinlan Scattering Experiments

Although stars approach the binary with a wide velocity distribution, consistent with the adopted σ, simulations are performed considering a unique value of the velocity at infinity (thus are **UNBOUND**) then accounting for a Maxwellian distribution Quinlan 1996.
This semplification leads to the estimate of the so called $H_1(v)$ first, which thus allows to obtian $H(\sigma)$ by integration over all possible velocities and weighted by MB distribution.
Therefore from each scattering experiment is measured the dimensionless energy change

$$C = \frac{a\Delta E_*}{G\mu} \tag{2.4}$$

and is then avereged over binary orientation (4π) and phase (2π).
By introducing the quantity

$$b_0^2 = \frac{2GM_{12}a}{v^2} \tag{2.5}$$

in order to work with the dimensionless impact parameter $x = b/b_0$, finally:

$$H_1 = \frac{1}{\pi b_0^2} \int db 2\pi b 4\pi \langle C \rangle = 8\pi \int dx \quad x\langle C \rangle \equiv 8\pi I_x(C). \tag{2.6}$$

Similarly, assuming a given velocity for scattering experiments, the eccentricity growth rate K is obtained from K_1.
The differentiation of

$$e^2 = 1 + \frac{2EL^2}{\mu(GM_{12})^2} \tag{2.7}$$

returns

$$\frac{\Delta e}{\Delta ln(1/a)} = \frac{1-e^2}{2e}\left[-2\left(\frac{E}{\Delta E}\right)\left(\frac{\Delta L_z}{L_z}\right) - 1\right] = \frac{1-e^2}{2e}(B-C), \tag{2.8}$$

where B is the dimensionless angular momentum change defined as

$$B \equiv -\frac{M_{12}}{m_*}\frac{\Delta L_z}{L_z}. \tag{2.9}$$

Finally, integrating over all possible impact parameter, angles and normalizing the result, K_1 can be computed as

$$K_1 = \frac{\int dx \quad xC(x)\langle\frac{\Delta e}{\Delta ln(1/a)}\rangle}{\int dx \quad xC(x)} = \frac{(1-e^2)}{2e}\left[\frac{I_x(B-C)}{I_x(C)}\right]. \tag{2.10}$$

NOTE:During three-body encounters, star-binary energy and angular momentum exchange ensures the conservation of both of them; thus the binary evolution can be inferred from stellar orbial properties evolution.

The initial setup of each scattering experimet is completely fixed by the following $d = 5$ uniformly distributed- within a proper interval- parameters :

1. b^2 rather than b because $t_{coll} \propto \lambda \propto \sigma^{-1} \propto b^{-2}$. Thus reproducing correctly the encounter rate.
2. cosine of the inclination: angle between binary $\mathbf{L}$ and $\mathbf{v}(-\infty)$.
3. longitude of ascending node: azimuthal angle of $\mathbf{v}(-\infty)$ in $\mathbf{L}$ equatorial plane
4. the argument of pericenter (point of closest approach): it fixes ellipse orientation over the orbital plane
5. the mean anomaly to specify BHs position along the ellipse at a given time.

Once these parameters are drawn and convinient coordinates are adopted, the numerical integration starts at binary-star separation $r = 50a$ and is stopped when the star leaves this sphere around the binary with posity energy, hence recording the main quantities of interest.
Systematic errors arise from approximations used in order to semplify the numerical simulation, for istance: assuming keplerian orbit outside that sphere, thus stopping at zero order term in multipole expansion.
Statistical errors in any average quantity scale, at large number of scatterings N, as $(ln \quad N)^d/N$ because quasi-random numbers are used.

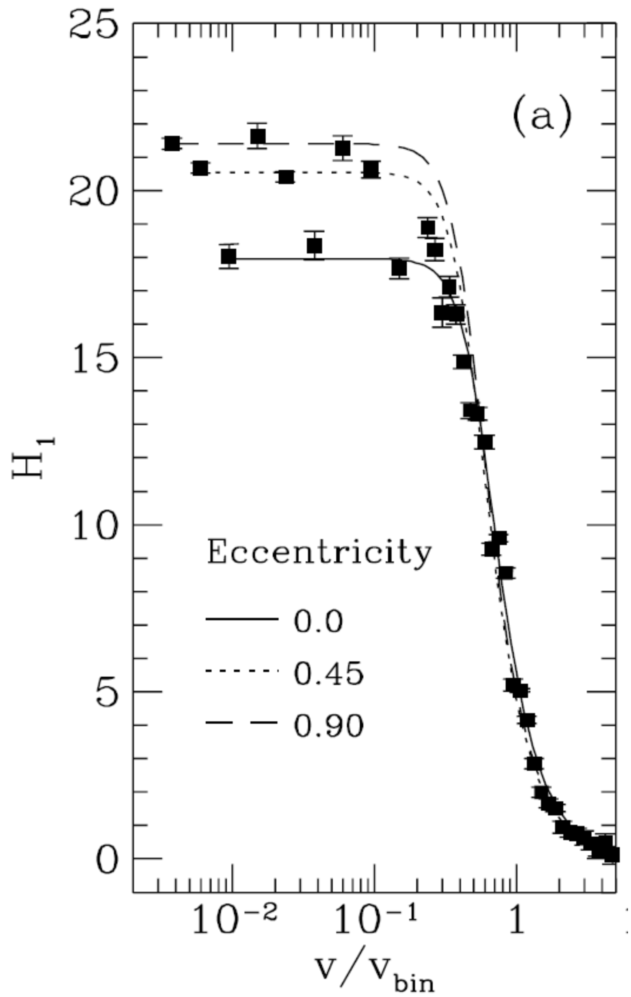


Figure 2.1: This plot shows that H_1 is almost indipendet of the eccentricity.

H_1 is measured for a wide range of masses ratio and eccentricity and fitted with the two free-parameters model

$$H_1(v) = H_0(1 + (v/w)^4)^{-1/2}. \tag{2.11}$$

NOTE: "This fits the data well at high and low velocities. It does not fit so well when H_1 first starts to decrease"

Similarly K_1 is derived as a function of "e" and fitted with the following three parameters model:

$$K_1(e) = e(1 - e^2)^{k_0}(k_1 + k_2 e) \tag{2.12}$$

NOTE: H_1 is a function of v only while K_1 on both e and v.

All free parameters are obtained for different masses ratio.

NOTE : in Sesana 2010, Fig 2.1 is reported as H vs a but this agrees with the functional form of H_1 and $V_{bin} = \sqrt{GM_{12}/a}$.
Therefore, the hardening rate for a circular binary have been used for all eccentricities.

2.1.3 Aplication to a galaxy model

Once stellar density ρ and velocity dispersion σ are specified, the binary BH is introduced at center of the system with a given initial eccentricity e_0 and separation a_0 such that $V_{bin} = \sigma/5$. The last corresponds to a condition about a_0 which is model indipendent and for the binary to be soft at beginning of the integration; a BH binary is indeed called hard if it hardens at a

constant rate, that is if $\sigma \ll \omega$ or equivalently if $V_{bin}/\sigma \gg (1 + m_1/m_2)^{1/2}$.
Peters (1964) equation for gravitational wave emission

$$\frac{da}{dt} = -\frac{64}{5}\frac{G^3 \mu m^2}{c^5 a^3}\frac{1}{(1-e^2)^{7/2}}\left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4\right) \tag{2.13}$$

$$\frac{de}{dt} = -\frac{304}{15}\frac{G^3 \mu m^2}{c^5 a^4}\frac{e}{(1-e^2)^{5/2}}\left(1 + \frac{121}{304}e^2\right) \tag{2.14}$$

are considered as well in order to compute da/dt and de/dt.
Despite dM_{ej}/dt is computed, a and e evolution can be assumed to occur in a fixed background whenever this is negligible; the ejected mass hardly exceeds a few times M_{12}.
Neverthless, the stellar mass ejection makes the binary hardening less and less efficient; once nearby stars are expelled, the hardening rate depends on the diffusion of stars back into the "Loss cone" due to two-body relaxation.
If stars removal is efficient, correction to the density profile must be applied assuming a given rate for the process above.
Fig.2.2 shows that the binary evolves mainly through three phases separated by two trashold destances:

1. $a_h = \frac{Gm_2}{4\sigma^2}$
2. a_{gw}

Quasi-circular binaries ($e_0 < 0.3$) remain circular with little e-growth before entering the GWs emission domain, while eccentricity increment is more important if $e_0 > 0.3$.

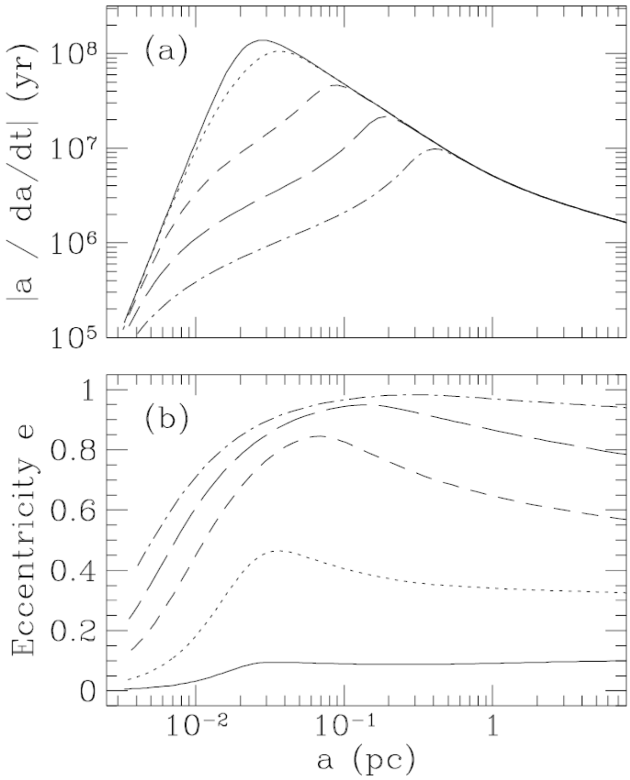


Figure 2.2: Eccentricity evolution. The solid line is the same as in Fig. 7(a); the other lines assume initial eccentricities of 0.3 (dotted), 0.5, 0.7, and 0.9 (dashed-dotted).

2.2 Sesana Hybrid Model

2.2.1 Sesana Scattering Experiments

This work Sesana, Haardt, et al. 2006 is aimed at unvieling by full three-body scattering experiments how a population of HVS form as a result of the interaction between a MBHB and ambient stars, unbound to the binary, drawn from a MB distribution . Other than the dynamical properties of these HVS, binary hardening, mass ejection rate and eccentricity growth rate (H, J, K) are tracked by numerical simulations.
Dynamical friction is efficient (especially in wet major mergers) in dragging two massive BHs towards the innermost region of the new formed galaxy; over the time scale t_{df} the binary forms and hardens through gravitational slingshot, hence ejecting low angular momentum stars passing close to the binary.
Following Q96 approach, the equations defining (H, J, K) are (2.1),(2.3) and (2.2).
At odds with Q96, where three-body scattering experiments are conducted for a fixed impact velocity, varying the binary separation and then averaging over the Maxwellian distribution, now the separation is fixed, i.e. $V_c = (GM_{12}/a)^{1/2} = const$, and the quantity v/V_c is sampled over four decades, corresponding to a sampling of about four decades in binary separation.
Each run is set by the inital configuration, corrseponding to a point in the parameter space (q, e_0); the combination of binary masses ratio $q = 1, 1/3, 1/9, 1/27, 1/81, 1/243$ and initial eccentricity $e_0 = 0.01, 0.3, 0.6, 0.9$ leads to 24 runs, involving $4 \cdot 10^6$ stars each.
The (Maxwellian averaged) rates H, J, and K derived from scattering experiments are fitted by the following functions

$$H = A(1 + a/a_0)^\gamma \tag{2.15}$$

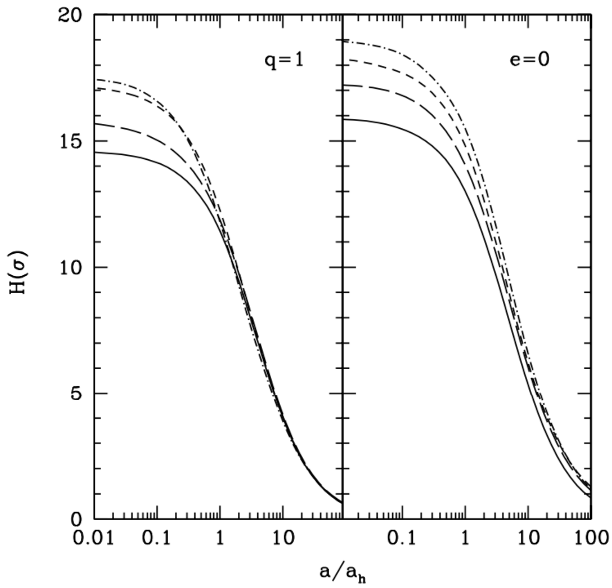


Figure 2.3: Binary hardening rate H (averaged over a Maxwellian velocity distribution) vs. a/a_h. Left: $e = 0, 0.3, 0.6, 0.9$ for $q = 1$. Right: q ¼ 1/3, 1/9, 1/27, and 1/81 for e ¼ 0. Line style as in Fig. 1

$$J = A(a/a_0)^\alpha[(1 + (a/a_0)^\beta]^\gamma \tag{2.16}$$

$$K = A(1 + a/a_0)^\gamma + B. \tag{2.17}$$

Figure 2.3 shows that the hardening rate is nearly constant when $a < a_h$, in perfect agreement with Q96 results. Therefore hard binary can be characterised by a costant hardening rate. Additionally, H decreases with q (asymmetric binaries harden faster) and increases with eccentricity (circular binaries harden more slowly).
As reported in Figure 2.4, K is close to zero at any binary separation for circular binary; if the binary is eccentric at a_h already, then K (and e) increases monotonically with binary shrinking. **NOTE**: in this paper the binary shrinking is studied taking into account gravitational slingshot on UNBOUND stars passing close to binary and ejected at higher velocity. In particular:

1. the ejection criterion relies on the stellar background model (here SIS stellar Bulge is considered)

2. The stellar background is assumed to be fixed; this approximation holds if the ejected mass does not exceed a trashold

3. Binary shrinking requires low angular momentum stars as they havo to pass closer and closer to the binary BHs for slingshot to be effective; however from $a < a_h$ the fraction of stars with these properties, populating the so called "Loss cone", is very small. Being depletd by three-body encounters, binary hardening when binary is hard depends on the efficiency of loss cone refilling.

Bound stars Sesana, Haardt, et al. 2008 ,Sesana 2010

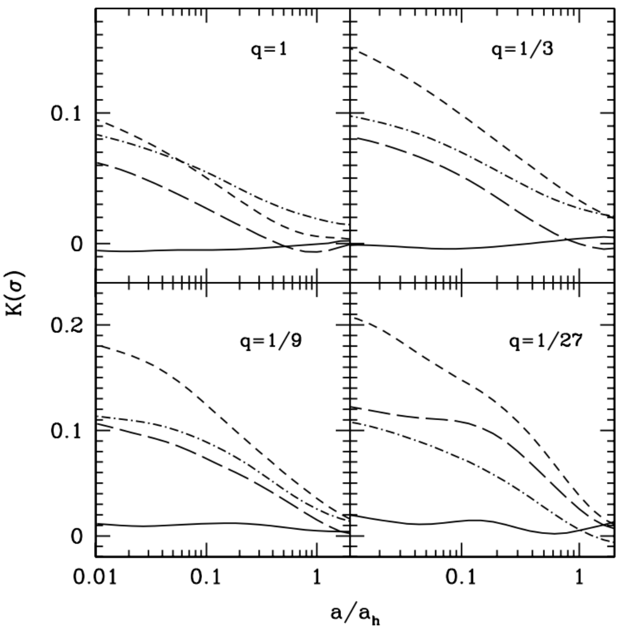


Figure 2.4: Eccentricity evolution. The solid line is the same as in Fig. 7(a); the other lines assume initial eccentricities of 0.3 (dotted), 0.5, 0.7, and 0.9 (dashed-dotted).

2.2.2 Sesana 2010 Results and Comparison

As confirmed by observations of galaxies hosting a dynamically evolved BH binary, stellar scouring by binary hardening seems to imprint an inner cusp to the original stellar background. Motivated by this emerging picture, Sesana Sesana 2010 adopted a stellar density profile matching a SIS outside the binary influence radius r_{inf} and a power low distribution inside; in formulae:

$$\begin{cases} \rho(r) = \rho_{inf}\left(\dfrac{r}{r_{inf}}\right)^{-\gamma} & if \quad r < r_{inf} \\ \\ \rho(r) = \dfrac{\sigma^2}{2\pi G r^2} & if \quad r > r_{inf} \end{cases}$$

Here σ indicates the radius-indipendent velocity dispersion of SIS model and it assumed to be a function of binary mass $M = M_1 + M_2$, such that $q = M_2/M_1 < 1$, through Tremaine et al. 2002

$$M_6 = 0.84\sigma_{70}^4. \tag{2.18}$$

Dynamical friction is efficient in dragging two BHs down to the separation a_0 at which the enclosed stellar mass in the binary is twice the mass of the secondary BH, i.e.

$$a_0 = (3-\gamma)\frac{GM}{\sigma^2}\left(\frac{q}{1+q}\right)^{1/(3-\gamma)} = r_{inf}\left(\frac{q}{1+q}\right)^{1/(3-\gamma)}, \tag{2.19}$$

and is adopted as starting point of the numerical integration.
Hereafter, three competing mechanisms are involved:

1. Bound cusp erosion:

$$\left.\frac{da}{dt}\right|_b = -\frac{2a^2}{GM_1M_2}\int_0^\infty \Delta\mathcal{E}\frac{d^2N_{\rm ej}}{da_\star dt}da_\star \tag{2.20}$$

$$\left.\frac{de}{dt}\right|_b = \int_0^\infty \Delta e\frac{d^2N_{\rm ej}}{da_\star dt}da_\star \tag{2.21}$$

2. Slingshot of unbound stars:

$$\left.\frac{da}{dt}\right|_u = -\frac{a^2 G\rho}{\sigma} H \tag{2.22}$$

$$\left.\frac{de}{dt}\right|_u = \frac{aG\rho}{\sigma} HK \tag{2.23}$$

3. Gravitational waves emission Peters and Mathews 1963:

$$\left.\frac{da}{dt}\right|_{\mathrm{GW}} = -\frac{64}{5}\frac{G^3}{c^5}\frac{M_1 M_2 M}{a^3(1-e^2)^{7/2}}\left(1+\frac{73}{24}e^2+\frac{37}{96}e^4\right) \equiv -\frac{64}{5}\frac{G^3}{c^5}\frac{M_1 M_2 M}{a^3}F(e) \tag{2.24}$$

$$\left.\frac{de}{dt}\right|_{\mathrm{GW}} = -\frac{304}{15}\frac{G^3}{c^5}\frac{M_1 M_2 M}{a^4(1-e^2)^{5/2}}e\left(1+\frac{121}{304}e^2\right) \tag{2.25}$$

Being indipendet of each other, by superposition, we can put the pieces together and the evolution of the binary obeys to

$$\frac{da}{dt} = \sum_i \left.\frac{da}{dt}\right|_i \tag{2.26}$$

$$\frac{de}{dt} = \sum_i \left.\frac{de}{dt}\right|_i . \tag{2.27}$$

Besides, an equal number of regimes can be introduced whereby one single mechanism is dominant over the others; lengthscale boundaries are a_0,

$$a_h \approx \frac{1}{4(3-\gamma)}\left(\frac{q}{1+q}\right)^{(2-\gamma)(3-\gamma)} \tag{2.28}$$

and

$$a_{GW} = 4\left[\frac{128\pi(3-\gamma)F(e)}{5H}\right]^{1/5}\frac{\sigma}{c}q^{-4/5}(1+q)^{3/5}a_h \tag{2.29}$$

respectively.
We recall that

$$\rho_{inf} = \frac{\sigma^2}{2\pi G r_{inf}^2} \tag{2.30}$$

and that setting $\rho = \rho_{inf}$ inside Equation (2.40) is equivalent to impose a full loss-cone outside the binary sphere of influence during the unbound phase; the effects due to different normalizations could be explored.
Sesana Sesana 2010 explored a relatively wide region of the phase space related to the initial configuration of the binary; here, we report the comparison between Sesana and our evolutionary track for $(q, \gamma, e_0, M) = (1, 1.5, 0.1, 10^5 M_\odot)$.
This kind of configuration, among the analyzed ones, is quite relevant to our studies on IMBH binaries at high z as they are expected to be more symmetric then their local universe cousins. As pointed out at the end of Section 1.3, a_h and r_{inf} are of the order of $\sim GM/\sigma^2$ and the bound cusp erosion process can be suppressed at first istance.
The estimate of the hardening radius, below which a_h is nearly costant, and the best fitting parameters in Table 1 Sesana, Haardt, et al. 2006 allows us to find

$$H = A(1 + a/a_0)^\gamma; \tag{2.31}$$

the function is shown in Figure 2.5.

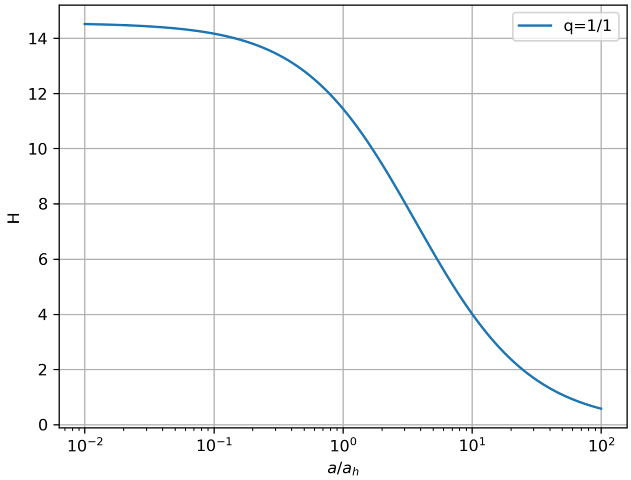


Figure 2.5: H as a function of a expressed in a_h units.

TABLE 1

HARDENING RATE

q	A	a_0	γ
1...........................	14.55	3.48	−0.95
1/3	15.82	4.18	−0.90
1/9	17.17	3.59	−0.79
1/27	18.15	3.32	−0.77
1/81	18.81	3.87	−0.82
1/243	19.16	4.16	−0.86

Figure 2.6: Note: a_0 is given in a_h units.

Similarly, by interpolation over the eccentricty grid of Table 3 Sesana, Haardt, et al. 2006 parameters, it is obtained

$$K = A(1 + a/a_0)^\gamma + B. \tag{2.32}$$

TABLE 3
ECCENTRICITY GROWTH RATE

e	A	a_0	γ	B
		$q = 1$		
0.15	0.037	0.339	−3.335	−0.012
0.3	0.075	0.151	−1.548	−0.008
0.45	0.105	0.088	−0.893	−0.005
0.6	0.121	0.090	−0.895	−0.008
0.75	0.134	0.064	−0.544	−0.006
0.9	0.082	0.085	−0.663	−0.004
		$q = 1/3$		
0.15	0.082	0.042	−0.168	−0.048
0.3	0.095	0.213	−1.152	−0.012
0.45	0.129	0.137	−0.655	−0.006
0.6	0.166	0.081	−0.546	−0.006
0.75	0.159	0.079	−0.497	−0.010
0.9	0.095	0.122	−0.716	−0.008
		$q = 1/9$		
0.15	0.051	0.385	−0.891	−0.011
0.3	0.111	0.307	−1.107	−0.007
0.45	0.172	0.526	−1.174	−0.016
0.6	0.181	0.251	−1.169	−0.007
0.75	0.179	0.195	−0.846	−0.004
0.9	0.117	0.400	−1.170	−0.001
		$q = 1/27$		
0.15	0.064	0.284	−1.206	0.021
0.3	0.143	1.033	−1.537	−0.021
0.45	0.212	0.722	−1.257	−0.022
0.6	0.216	0.430	−1.163	−0.014
0.75	0.173	0.771	−1.934	−0.014
0.9	0.129	0.329	−1.125	−0.020

Figure 2.7: Note: a_0 is given in a_h units.

Figure 2.8,2.9, 2.10 and 2.11 report the results of this mathematical operation.

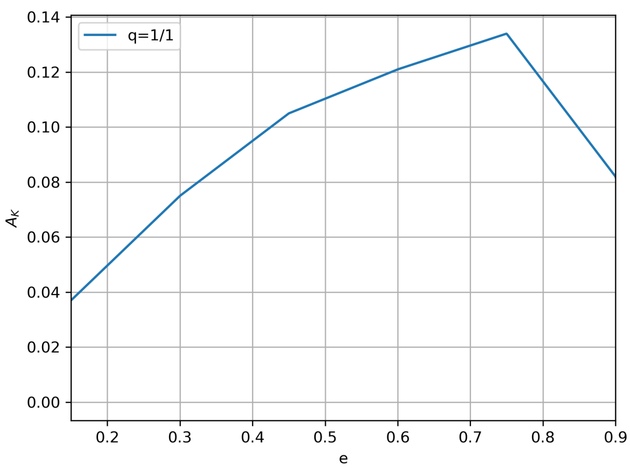


Figure 2.8: A as a function of e derived by interpolation of grid points.

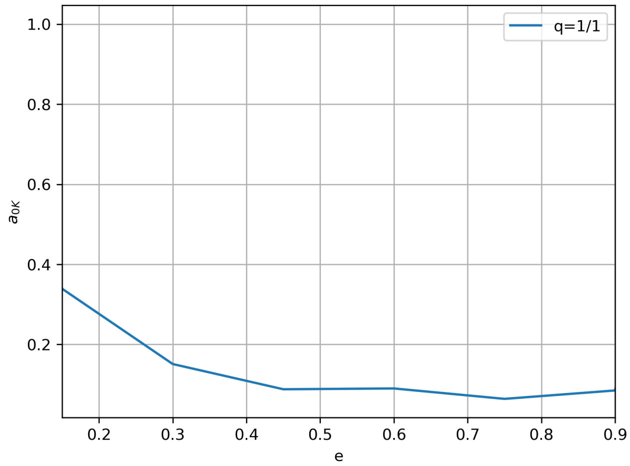


Figure 2.9: A as a function of e derived by interpolation of grid points.

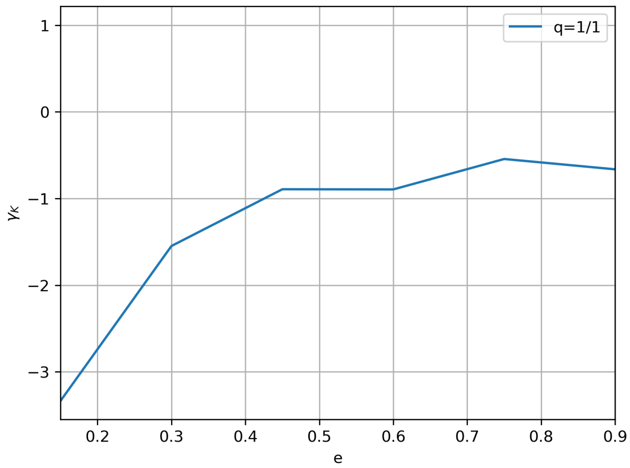


Figure 2.10: γ as a function of e derived by interpolation of grid points.

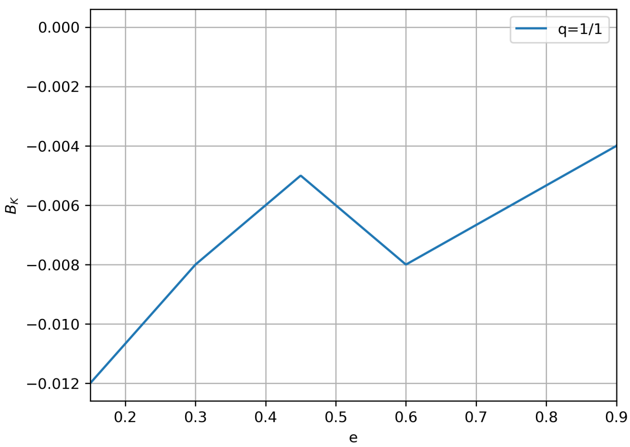


Figure 2.11: B as a function of e derived by interpolation of grid points.

Therefore, at odds with H, K is a function of (a, e); we list below the plots of the 1-D functions $K(a) = K(a, e_0)$ and $K(e) = K(a_h, e)$.

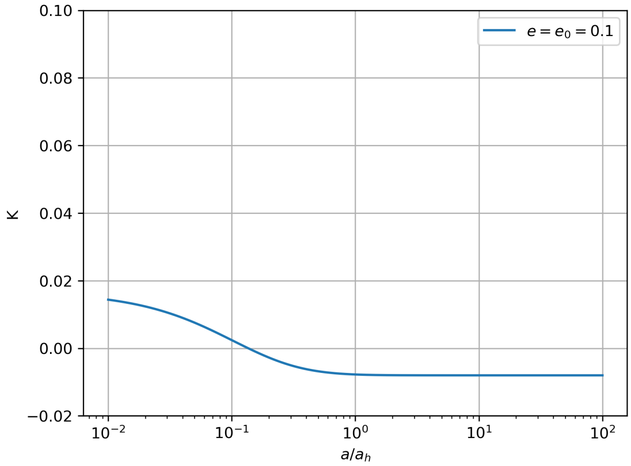


Figure 2.12

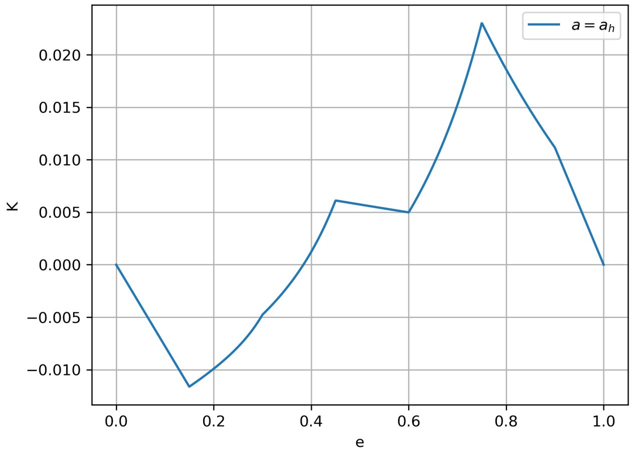


Figure 2.13

We anticipate that Equations (2.26) and (2.27) will be integrated with the help of RK(4) and dK/de is needed as well; Figure 2.14 depicts the step-like function.

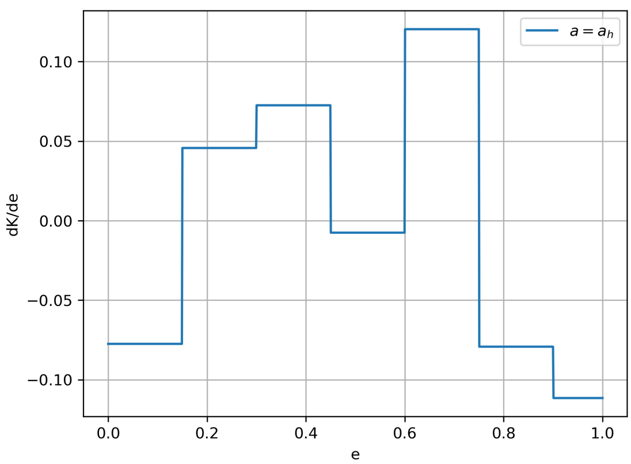


Figure 2.14

The time step is adjusted at each iteration according to the prescription

$$\Delta t \leq min \left\{ \frac{2.79}{|\lambda_i|} \right\} \qquad \forall i = 1, N \tag{2.33}$$

where λ_i are the Jacobian $\mathbf{J}$ eigenvalues. The elements of this matrix are:

$$J_{11} = \left.\frac{da}{dt}\right|_u \left(\frac{2}{a} + \frac{\gamma}{a + a_0 a_h} \right) - \left.\frac{da}{dt}\right|_{gw} \left(\frac{3}{a} \right) \tag{2.34}$$

$$J_{12} = \left.\frac{da}{dt}\right|_{GW} \left(\frac{7e}{1-e^2} + \frac{73e/12 + 37e^3/24}{1 + 73e^2/24 + 37e^4/96} \right) \tag{2.35}$$

$$J_{21} = \left.\frac{de}{dt}\right|_u \left(\frac{1}{a} + \frac{\gamma_H}{a + a_{0,H} a_h} + \frac{\gamma_K}{a + a_{0,K} a_h} \right) - \left.\frac{de}{dt}\right|_{GW} \left(\frac{4}{a} \right) \tag{2.36}$$

$$J_{22} = \left.\frac{de}{dt}\right|_u \left(\frac{\partial \ln K}{\partial e} \right) + \left.\frac{de}{dt}\right|_{GW} \left(\frac{5e}{1-e^2} + \frac{1 + 363e^2/304}{e + 121e^3/304} \right) \tag{2.37}$$

In order to compare Sesana predictions with our model we set (t_0, a_0, e_0) for the unbound phase to be effective and selecting these values directely from Sesana plot: $(t_0, a_0, e_0) = (12Myr, 0.05 \cdot a_0, 0.131)$.

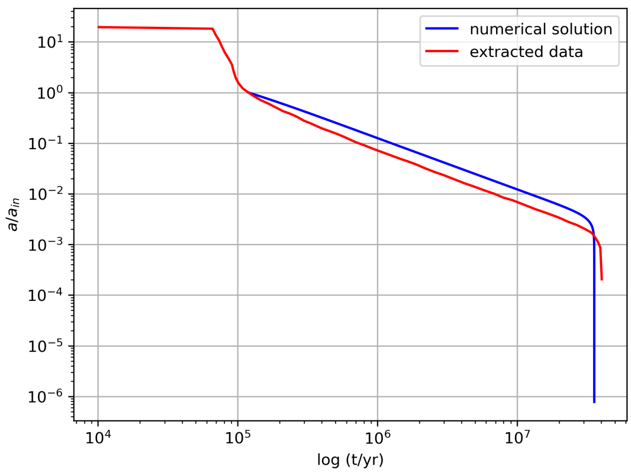


Figure 2.15

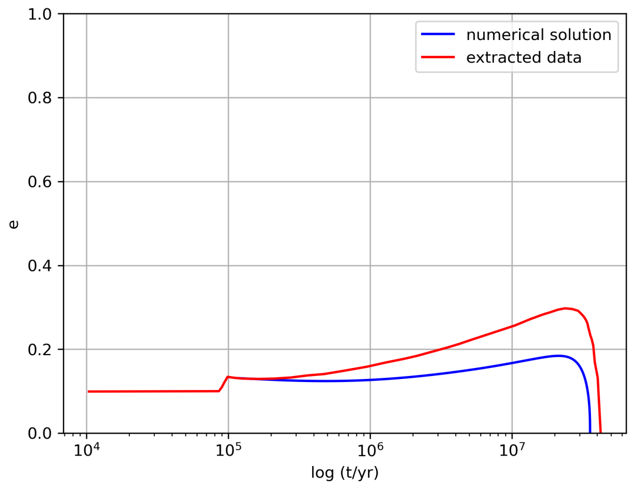


Figure 2.16

Figure 2.15 and 2.16 indicate a good agreement between our model and Sesana one. In the first plot a decreases more slowly than Sesana prediction as we are negletting the bound cusp erosion contribute, which hardens the binary even when the unbond phase takes over at $\sim a_h$. On the other hand eccentricity reaches a lower peak compared to the reference model because of bound cusp erosion leads to e growth as well.
Furthermore, we test the feedback of the various shrinking mechanisms by assuming them working in isoletion, hence suppressing all the others.

Figure 2.17 and 2.18 refers to the evolution under the effect of slinghot of unbound stars; as the initial separation is below a_h we can appreciate the expected linear decay of a and the overall eccentricity growth.

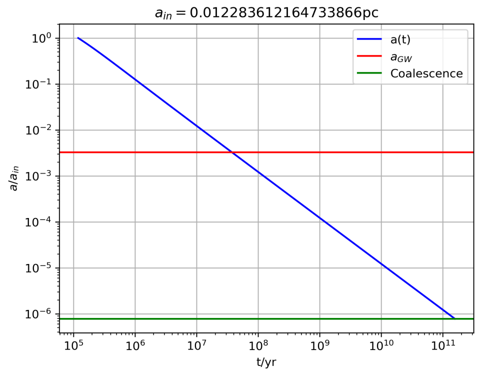


Figure 2.17

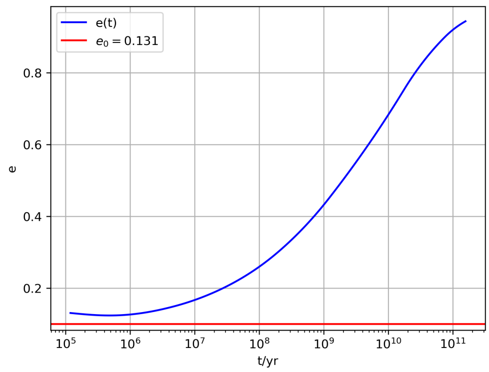


Figure 2.18

Figure 2.19 and 2.20 refers to the evolution under GWs production; this becomes effective only at very close separation and causes both the fast binary coalescence once effective and binary circularization. The coalescence time agrees perfectly to the theoretical prediction of Equation 1.105

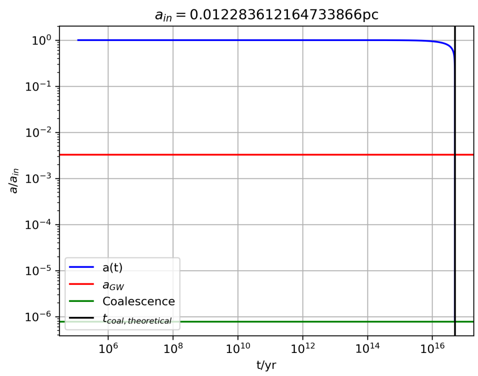


Figure 2.19

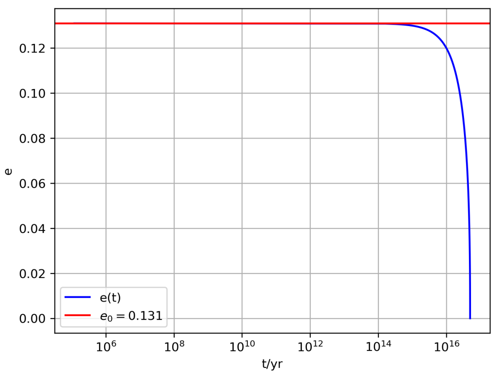


Figure 2.20

The dataset used above help us to construct another curve, precisely $(e(t, a(t)))$ which can be compared to the close form solution provided by Equation (1.102) as shown in Figure 2.21 and 2.22.

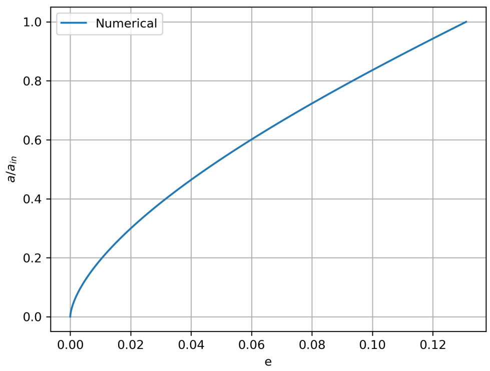


Figure 2.21

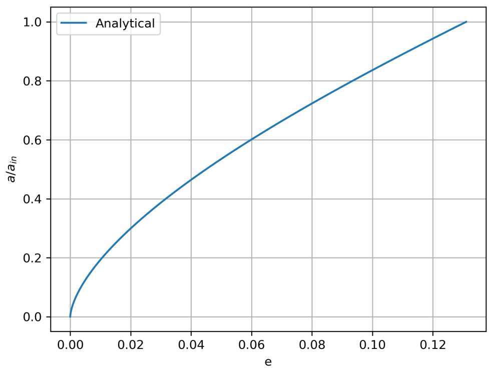


Figure 2.22

Below, the Figure 2.23 gives us the possibility of visualizing the relative error of the numerical solution with respect to the analytical one; it turnsout to be within one part over 10^{12}.

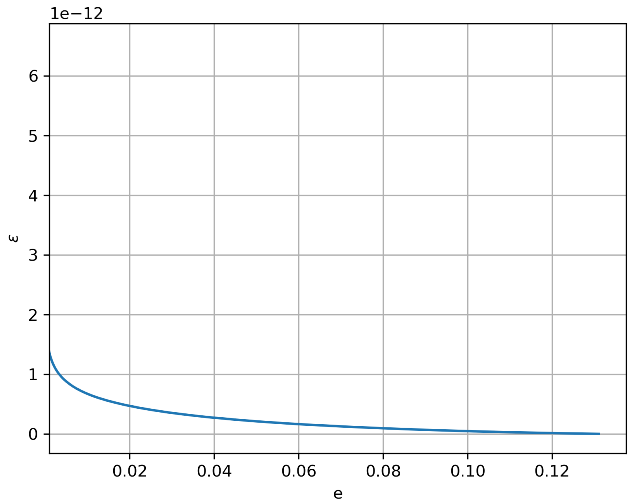


Figure 2.23

2.3 NGC 205 IMBH Binary evolution

2.3.1 NGC 205 Hybrid Model Evolution

NGC 205 is a nearby dwarf galaxy with a NSC residing at its innermost region; it is belived to host an IMBH binary, whose estimated mass is $\sim 10^4 M_\odot$.
The 2-D surface brightness profile of the dwarf, both the bulge and NSC treated as indipendent stellar components, is translated into 3-D Sersic density profile using Prugniel & Simien Prugniel and Simien 1997:

$$\rho(r) = \rho_0 \left(\frac{r}{R_e}\right)^{-p} e^{-b\left(\frac{r}{R_e}\right)^{1/n}}. \tag{2.38}$$

Here, ρ_0 is the density normalization unambiguously set by total galaxy mass, effective radius and Sersic index n; from Terzić and Graham 2005:

$$\rho_0 = \frac{M_*}{4\pi R_e^3} \frac{b^{n(3-p)}}{n\,\Gamma[n(3-p)]}. \tag{2.39}$$

The parameters p and b can be expressed as a function of n as follows: $p = 1.0 - 0.6097/n + 0.05563/n^2$, and $b \approx 2n - 1/3 + 0.009876/n$, for $0.5 < n < 10$.
Observations constrain the quantites (M, R_{eff}, n) for both bulge and NSC as reported in Figure (2.24) of Khan and Holley-Bockelmann 2021.

Component	n	r_{eff}(pc)	$M_\star$ (10^7 M$_\odot$)	a_2/a_1
IMBH	–	0.14	0.004	–
NSC	1.6	1.3	0.18	0.95
Bulge	1.4	516	97.2	0.90

Figure 2.24: Column 1: Galaxy component. Column 2: Sersic index. Column 3: Effective radius of the galaxy component (note: for the IMBH, we quote the influence radius instead). Column 4: Stellar mass. Column 5: 2D axial ratio as measured by GALFIT.

Numerically it is found

- Bulge: $\rho_0 = 4.80 \cdot 10^5 M_\odot/pc^3$
- NSC: $\rho_0 = 2.95 M_\odot/pc^3$

and NGC 205 density profile is reconstructed in Figure (2.25).

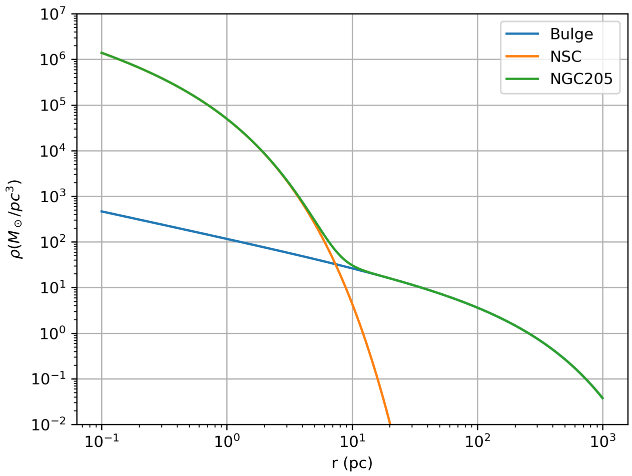


Figure 2.25: NGC 205 bulge, NSC and total mass density profile of stellar distribution.

Adopting the same prescription of Khan and Holley-Bockelmann 2021 and Holley-Bockelmann et al. 2025, a symmetric IMBH binary of masses $M_1 = M_2 = 4 \cdot 10^4 M_\odot$ is introduced into NGC 205 core and its evolution, due to stellar and GWs hardening, is tracked combining 3-body scattering experiment hybrid model Quinlan 1996,Sesana, Haardt, et al. 2006 and Peters and Mathews 1963 equations.
It is quite relevant to point out that in this astrophysical system the hardening due to bound stars Sesana, Haardt, et al. 2008 is negligible in light of the symmetry of the binary and the arguments exposed at the end of Sec.(1.3).
Therefore, combining the set of equations related to Unbound evolution

$$\frac{da}{dt}|_u = -\frac{a^2 G \rho_{\text{inf}}}{\sigma} H \tag{2.40}$$

$$\frac{de}{dt}|_u = \frac{a G \rho_{\text{inf}}}{\sigma} H K \tag{2.41}$$

and the equations describing GWs driven hardening 1.99 1.100, the set of differential equations must be solved reads

$$\frac{da}{dt} = \frac{da}{dt}|_u + \frac{da}{dt}|_{gw} \tag{2.42}$$

$$\frac{de}{dt} = \frac{de}{dt}|_u + \frac{de}{dt}|_{gw} \tag{2.43}$$

Although the original formulation of the hybrid model relies on a SIS profile, with very peculiar features (i.e. costant velocity dispersion), Sesana Sesana and Khan 2015 showed how to use it for a more general stellar background. The vast majority of stars, unbound to the binary, refilling the loss cone (i.e. having pericenter $\sim a$) come from a region that is just beyond the binary influence radius, hence r_{inf} is the relevant physical scale of the model and σ appearing in Equations 2.40 and 2.41, which is in general a function of the radius for spherically symmetric systems, must be computed at the influence radius (here defined as the radius enclosing stellar mass that is twice the binary one).
Apart from setting the inital point in the (a, e) space, we need to determine several quantities; we start with binary influence radius and $\sigma_{inf} = \sigma(r_{inf})$; they both require the knowledge of the cumulative mass $M(< r)$ associated to the Sersic profile. It results from the integration of $\rho(r)$ and admits analytic solution Terzić and Graham 2005

$$M(< R) = M_{\rm tot} \frac{\gamma\left(2n,\ b_n \left(\frac{R}{R_e}\right)^{1/n}\right)}{\Gamma(2n)}. \tag{2.44}$$

The root of the equation

$$M(< r_{inf}) - 2(M_1 + M_2) = 0 \tag{2.45}$$

is obtained numerically by bisection and holds $r_{inf} = 0.47pc$.
On the other hand, Jeans equation provides a way to compute the velocity dispersion and in case of spherical distributions it greatly simpifies to

$$\sigma^2(r) = \frac{1}{\rho(r)} \int_r^{\infty} \rho(s) \frac{GM(< s)}{s^2} ds. \tag{2.46}$$

The integral leads to σ_{inf} =32.6km/s. As both $\sigma(r)$ and the circular speed $V_c(r) = \sqrt{GM(< r)/r}$ depend on $M(< r)$, previously obtained, we can derive them and Figure 2.26 displays their ratio as afunction of the radius.

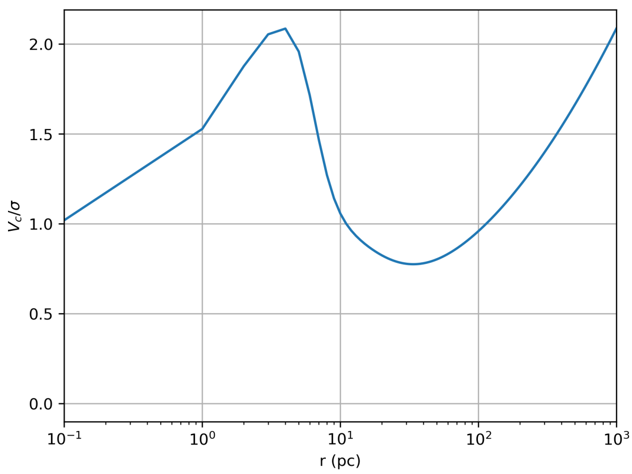


Figure 2.26: NGC 205 V_c/σ vs r

The last quantity we have to specify in order to implement the hybrid model is the binary hardening radius a_h; indeed Sesana Sesana, Haardt, et al. 2006 found out from 3-body scattering experiments that H and K functions can be fitted to

$$H = A(1 + a/a_0)^{\gamma} \tag{2.47}$$

and

$$K = A(1 + a/a_0)^{\gamma} + B \tag{2.48}$$

The parameters of the fits to H and J are listed in Tables 1 and 2 of Sesana, Haardt, et al. 2006 for different binary mass ratios and eccentricity; there, a_0 is given in a_h units, thus forcing us to determine it. In literature several definitions have been proposed; hereafter, we will be using

$$a_h \approx \frac{GM_2}{4\sigma^2}, \tag{2.49}$$

unless otherwise stated. This definition leads to $a_h \approx 0.04pc$.
We can finally proceed and numerically integrate the set of equations 2.42 and 2.43 by means of R-K(4). For the temporal evolution to be properly resolved from the initial configuration to the coalescence of the binary, the time step adjusts itself at each iteration, thus taking care of whether the dominant timescale is set by stellar hardening or GWs hardening at that time.
By defining

$$\Delta t_i = \epsilon \cdot min\left(\frac{a}{|da/dt|_u}, \frac{a}{|da/dt|_{gw}}\right), \tag{2.50}$$

where $\epsilon \sim 10^{-3-4}$, and the initial conditions $(a_0, e_0) = (15pc, 0.7)$, the integration returns $a(t)$ and $e(t)$.
They are shown in Figure 2.27 and 2.28.

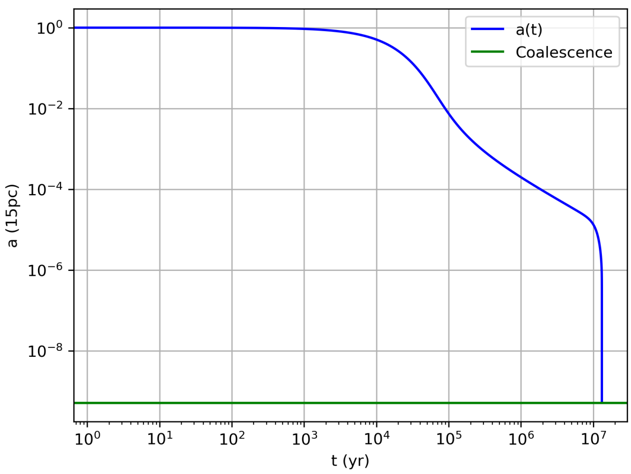


Figure 2.27: NGC 205 a vs t

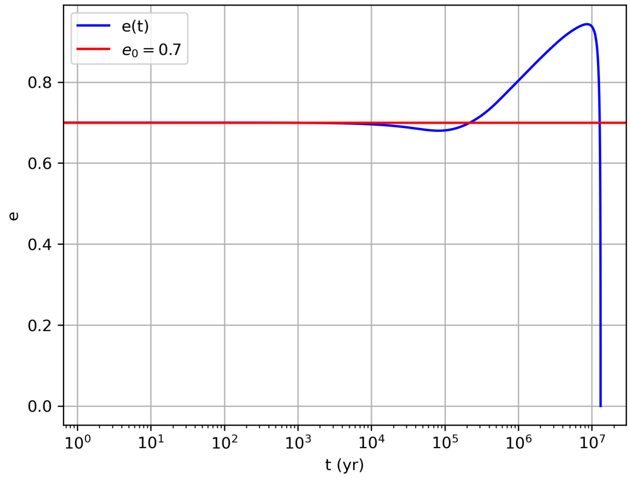


Figure 2.28: NGC 205 e vs t

The estimeted coalescence time is $t_{coal} = 13.15 Myr$; it is of the same order of magnitude ofKhan and Holley-Bockelmann 2021 and Holley-Bockelmann et al. 2025 predictions of $t_{coal} = 74 Myr$.
NOTE: Khan estimate comes from N-body simulations and in the special case of NGC 205 a costant value of eccentricity $e = 0.39$ is assumed for the late evolution of the binary.
Assuming a reliable time (universal time) at which the binary is in the configuration (a_0, e_0) it might be interesting to get an estimate of the corresponding z_{coal} from

$$t(z) = \int_z^{\infty} \frac{dz'}{(1+z')H(z')} \tag{2.51}$$

and compare it with LISA upper limit ($z \sim 20$), hence addressing the problem of a possible detection.
If the binary is in the initial configuration at $z \approx 10$, it will coalesce at $z \approx 9.4$.

2.3.2 Cosmic Size Evolution Down to NGC 205

Here, we explore the importance of the existence of NSC to the binary evolution and coalescence time. A such structure is expected to drastically reduce the time taken by system to merge as a consequence of the larger number of stars capable of passing close to the binary.
In this spirit, we will be referring to the same parameters table a procedure of the prevois SubSection but the NSC which is artificially removed from the dwarf galaxy; hence, the only stellar background affecting binary evolution is the stellar bulge.
By repeating the same analysis and setting $(t_0, a_0, e_0) = (0yr, 15pc, 0.7)$, we show Figure 2.29 and 2.30.

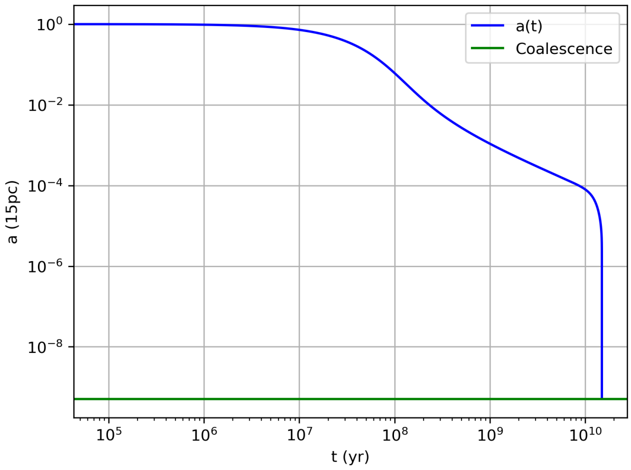


Figure 2.29

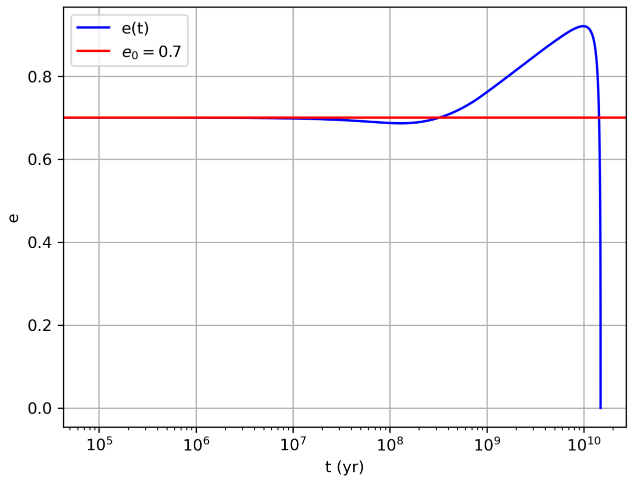


Figure 2.30

Now $t_{coal} = 14.93 Gyr$, almost three orders of magnitude longer than the NSC scenario. Following Ono cosmic size evolution relation $R_{eff}(z) = R_{eff}(z = 0)(1+z)^{-1.22}$ Ono et al. 2023, we take NGC 205 as a $z = 0$ reference model and we discuss the evolution of the same IMBH binary inside its plausible progenitors distributed between $z = 1$ and $z = 20$. All galaxies share same bulge mass and Sersic index; density profiles differ from each other only because of the different effective radius as diplayed in Figure 2.31.

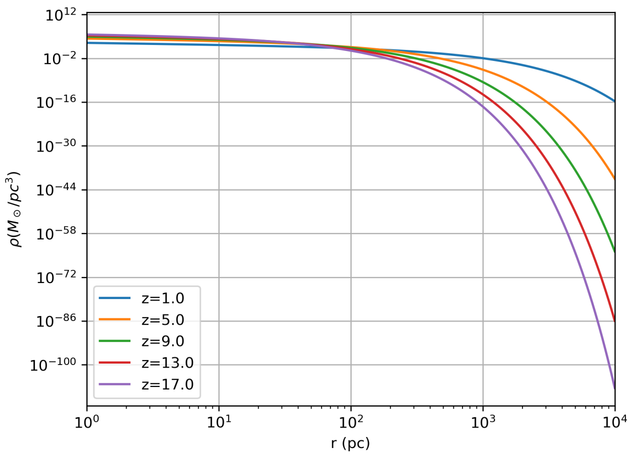


Figure 2.31

High redshift galaxies tend to be more compact than low redshift counterparts. Hardening radii are easly derived from these profiles according to Equation 2.49 and a comparison between different hosts H and K functions- see Figure 2.32 and 2.33- suggest that IMBH binary becomes "hard" at lower separation if embedded in a denser environment.

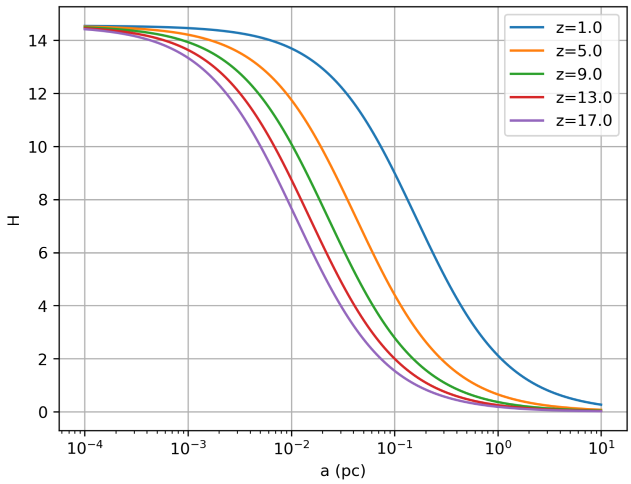


Figure 2.32

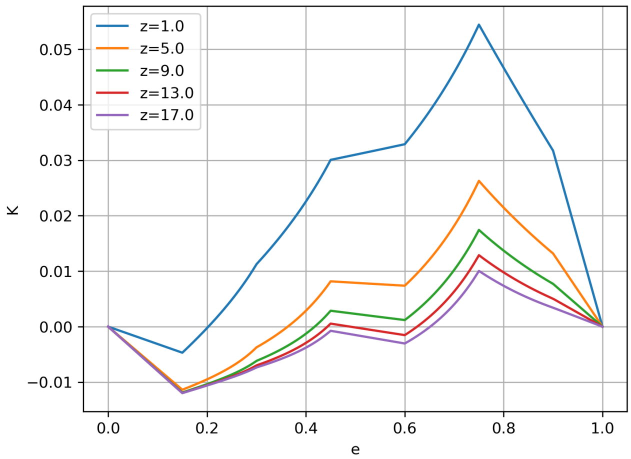


Figure 2.33: This family of functions is obtained evaluating $K(a, e)$ of each galaxy at their mean hardening radius.

We present in Figure 2.34 and 2.35 the evolutionary tracks of a sample of galaxies obtained by applying RK(4) to the Equations (2.42) and (2.43) and initial condition $(t, a_0, e_0) = (0yr, 15pc, 0.7)$.

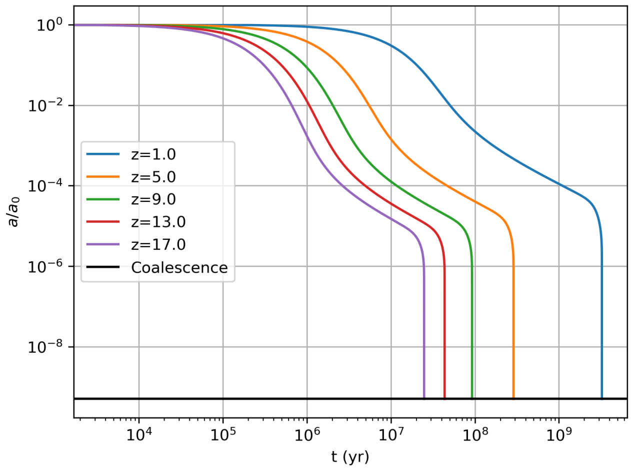


Figure 2.34

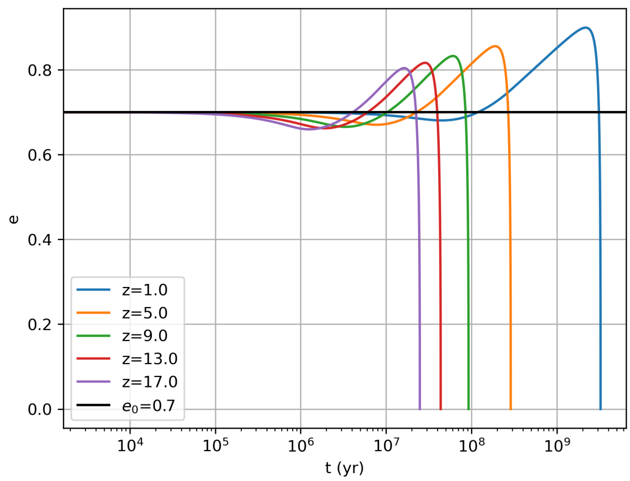


Figure 2.35

We note that higher compactness tends to compensate the effects of NSC removal in terms of the coalescence time.
Lastly, we modifiy the initial time t_0 of each run, turning it into $t(z)$, thus focusing on the possibility of coalescence within the present time and consequently a potential GWs observation.

Table 2.1: Galaxies coalescence time grid (Gyr)

z	t_{coal} (Gyr)
1	3.24728387
2	1.334011
3	0.7072763
4	0.43164087
5	0.28805534
6	0.20450559
7	0.15193439
8	0.11686942
9	0.09240003
10	0.07469761
11	0.06150781
12	0.05143637
13	0.04358488
14	0.03735426
15	0.03233322
16	0.02823207
17	0.02484224
18	0.02201059
19	0.01962278
20	0.01759202

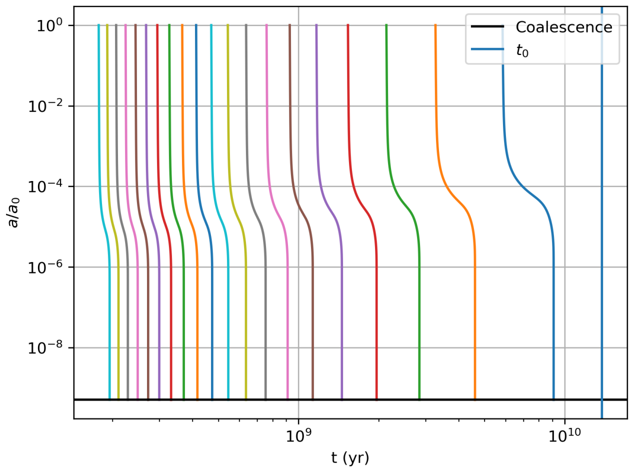


Figure 2.36

Therefore, forming in tha past at $z > 0$, they all merge before the present time.

3

N-BODY

Within t_{2b}, usually much shorter than the age of the universe for common galaxies, granularity is unimportant, collisions are negligible and stars can be described as particles moving in a smooth potential generated by a continous ρ. Hence, stars will have a DF satisfying the CBE:

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \dot{\mathbf{w}} \cdot \frac{\partial f}{\partial \mathbf{w}} = \frac{\partial f}{\partial t} + [f, H] = 0 \tag{3.1}$$

In cartesian coordinates the Hamiltonian is

$$H(\mathbf{x}, \mathbf{v}) = \frac{1}{2}v^2 + \Phi(\mathbf{x}) \tag{3.2}$$

and the CBE becomes

$$0 = \frac{\partial f}{\partial t} + \mathbf{v} \cdot \frac{\partial f}{\partial \mathbf{x}} - \frac{\partial \Phi}{\partial \mathbf{x}} \cdot \frac{\partial f}{\partial \mathbf{v}} \tag{3.3}$$

Several times beyond t_{2b} the system enters the collisional regime and first order corrections to the CBE are needed; the Fokker-Planck equation is an example of this:

$$\frac{Df}{Dt} = C[f] \tag{3.4}$$

where simplifications on the collision operator $C[f]$ can be done.
In this scenario f is no longer conserved along the orbit of the potential Φ and diffusion in the PS occurs, both E and L of a single star can change due to collisions, at odds with the CBE regime where, for istance $DH/Dt = 0$. It is used in the loss cone problem.
In order to exploit the properties of the CBE it is usefull to introduce the velocity moments, which will appear inside the Jeans equations:

$$\overline{v_i}(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathcal{R}}^{3} d^3\mathbf{v} f v_i \tag{3.5}$$

$$\sigma_{ij}^2(\mathbf{x}, t) \equiv \frac{1}{\rho(\mathbf{x}, t)} \int_{\mathcal{R}}^{3} d^3\mathbf{v} f (v_i - \overline{v_i})(v_j - \overline{v_j}) = \overline{v_i v_j} - \overline{v_i}\,\overline{v_j} \tag{3.6}$$

A function $I(\mathbf{x}, \mathbf{v})$ is an integral of motion if and only if

$$\frac{D}{Dt}[I(\mathbf{x}(t), \mathbf{v}(t))] = 0 \tag{3.7}$$

along any orbit of the potential Φ.
Jeans Theorem: Any steady-state solution of the collisionless Boltzmann equation depends on

the phase-space coordinates only through integrals of motion in the given potential, and any function of the integrals yields a steady-state solution of the collisionless Boltzmann equation. In a steady-state potential the Hamiltonian is an integral of motion; therefore f will be any non-negative function of H.
Ergodic DFs: they are of the kind $f = f(H)$.
Equilibrium stellar systems have $\overline{v_i} = 0$ and isotropic velocity-dispersion tensor (being symmetric it can be diagonalized at any point in a proper orthogonal set of axes), i.e $\sigma_{ij}^2 = \sigma^2\delta_{ij}$. Thus every system with an ergodic DF is isotropic.
If the potential Φ is time-indipendent and spherically symmetric, then $\mathbf{L}$ appears as three isolating integral of motions but DF will depend on them only through $|\mathbf{L}| = L$, i.e. $f = f(H, L)$.Adopting spherical coordinates and splitting $v^2 = v_r^2 + v_t^2$, it holds

$$\overline{v_r} = 0 \tag{3.8}$$

$$\overline{\mathbf{v_t}} = \mathbf{0} \tag{3.9}$$

and in general σ_{ij} is diagonal with $\sigma_r^2 \neq \sigma_\theta^2 = \sigma_\phi^2$.
Recovering the DF from ρ can be done by Eddington's formula, which applies to spherically symmetric, globally isotropic stellar systems,

$$f(\varepsilon) = \frac{1}{\sqrt{8}\pi^2}\frac{d}{d\varepsilon}\int_{\varepsilon_t}^{\varepsilon}\frac{d\rho}{d\Psi}\frac{d\Psi}{\sqrt{\varepsilon - \Psi}} \tag{3.10}$$

with $\varepsilon = -H$ and $\Psi = -\Phi$ being the relative energy and potential respectively and ε_t the truncation energy of the system.
Given the density profile, IC for N-body simulations are generated by OpOpgadget software by sampling uniformly the internal mass profile and the potential produced by the smooth profile ρ.
The Sersic profile introduced by Prugniel and Simien (1997) has (with standard substitution $Z = b\left(\frac{r}{R_e}\right)^{1/n}$)

$$M(<r) = 4\pi\rho_0 R_e^3 n b^{n(p-3)}\gamma\left[n(3-p), b\left(\frac{r}{R_e}\right)^{1/n}\right] = M_*\frac{\gamma\left[n(3-p), b\left(\frac{r}{R_e}\right)^{1/n}\right]}{\Gamma[n(3-p)]} \tag{3.11}$$

with

$$\Gamma(a) = \int_0^\infty dte^{-t}t^{a-1} \tag{3.12}$$

the "Complete gamma function",

$$\Gamma(a, x) = \int_x^\infty dt\, e^{-t}t^{a-1} \tag{3.13}$$

the Upper incomplete gamma function and

$$\gamma(a, x) = \int_0^x dt\, e^{-t}t^{a-1} \tag{3.14}$$

the Lower incomplete gamma function. By definition follows:

$$\Gamma(a) = \Gamma(a, x) + \gamma(a, x) \tag{3.15}$$

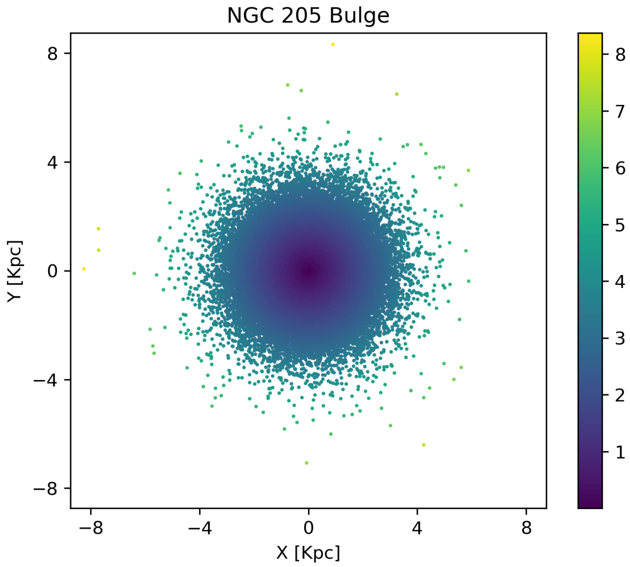


Figure 3.1: aaa

The spatial distribution is spherical and untruncated but the mass profile converges to the finite value M_*, so the potential can be found from

$$\Phi(r) = \Phi(r_0) + \frac{GM(<r_0)}{r_0} - \frac{GM(<r)}{r} - 4\pi G \int_r^{r_0} dt\, t\rho(t) \tag{3.16}$$

Setting $r_0 \to \infty$ and $\Phi(r_0) = 0$, it becomes

$$\Phi(r) = -\frac{GM(<r)}{r} - 4\pi G \int_r^{\infty} dt\, t\rho(t) \tag{3.17}$$

and computation returns

$$\Phi(r) = -4\pi G\rho_0 R_e^2 n\, b^{n(p-2)} \left\{ \frac{b^{-n}}{r/R_e} \gamma[n(3-p), Z] + \Gamma[n(2-p), Z] \right\} \tag{3.18}$$

or equivalently

$$\Phi(r) = -\frac{GM_*}{r} \left\{ \frac{\gamma[n(3-p), Z]}{\Gamma[n(3-p)]} + Z^n \frac{\Gamma[n(2-p), Z]}{\Gamma[n(3-p)]} \right\} \tag{3.19}$$

The generation of IC for the dwarf galaxy NGC 2O5, hereafter taken as a $z = 0$ nucleated dwarf galaxy, is carried by OpOpgadget; the Sersic profile defined in this software is based on the one analyzed by Lima Neto et al. (1999), i.e.

$$\rho(r) = \rho_0 \left(\frac{r}{a}\right)^{-p} e^{-(r/a)^{\nu}}, \tag{3.20}$$

where $n = \nu^{-1}$, a is the Sersic profile scalelength and no $b(n)$ is present. The same model can be described through several combinations of indipendent parameters; here, in order to use the effective radii values obtained from sky survy, we need to use the analytical approximation

$$\ln R_{eff}/a = \frac{0.6950 - \ln \nu}{\nu} - 0.1789 \tag{3.21}$$

COM data:

- position:

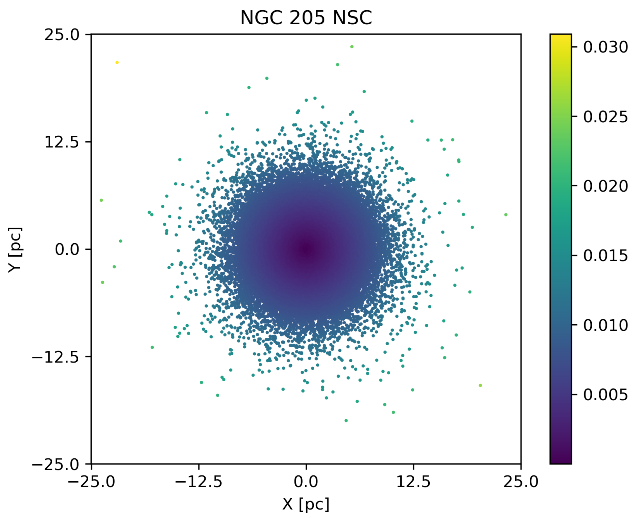


Figure 3.2: aaa

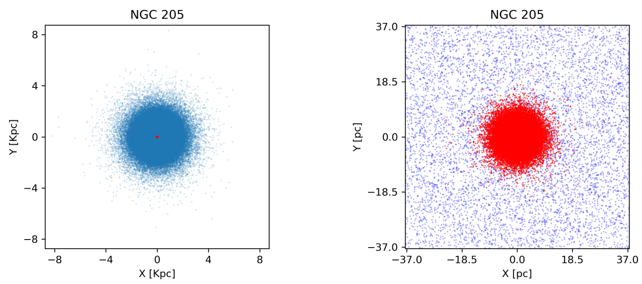


Figure 3.3: aaa

- velocity:

To test the stability of the Nbody generated IC, we let the system free to evolve over several tens of the dynamical time t_{dyn} at half-mass radius r_{half}; by definition, the former is

$$t_{dyn} = \frac{1}{\sqrt{G\bar{\rho}}} \tag{3.22}$$

and the latter reads, for a spherical system,

$$\bar{\rho} = \frac{M/2}{\frac{4}{3}\pi r_{half}^3} \tag{3.23}$$

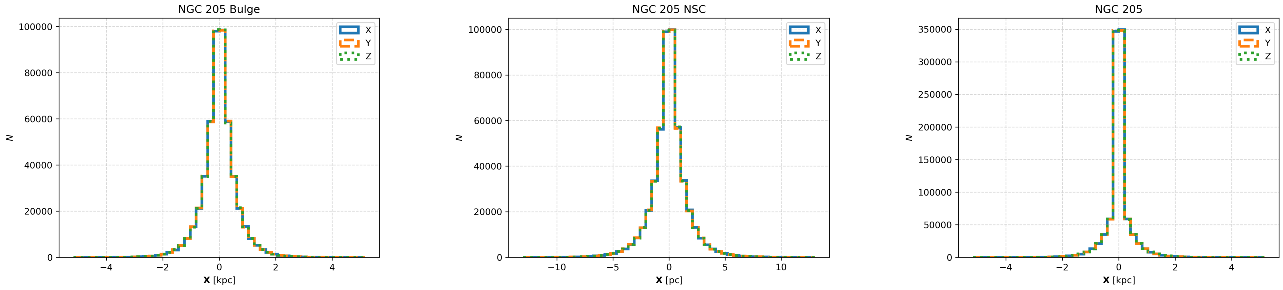
Numerically is found $r_{half} = ... = ...R_{eff}$.

Yet, we get $t_{dyn} = ...$, which agrees within $x\%$ the estimate obtained from the Analysis tool provided by OpOpgadget, which measures this quantity directely from the generated model. Waiting for evolution with Ketju.
Additionaly, the virial ratio Q has been computed as well as it gives a first indication of whether the model is virialized, i.e. at equilibrium, or expansion/contraction operate within several dynamical times. In S_{CM}, it is defined as

$$Q = \frac{2K}{|U|} \tag{3.24}$$

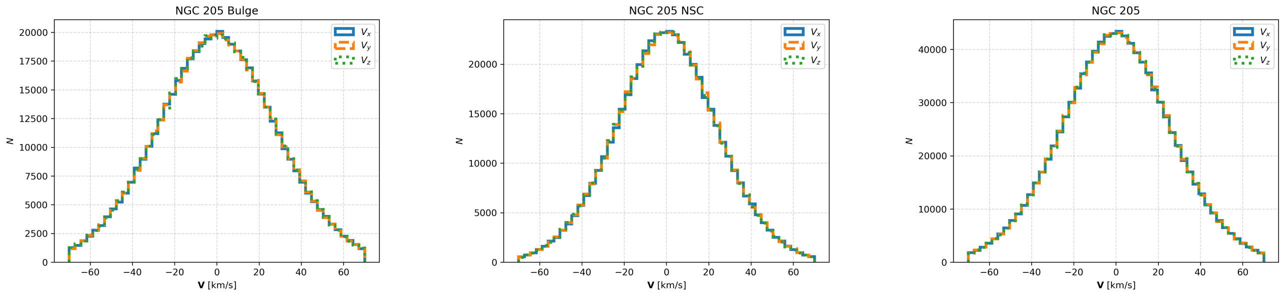
where

$$K = \sum_{i=1}^{N} \frac{1}{2} m_i v_i^2 \tag{3.25}$$



(a) A (b) B (c) C

Figure 3.4: aaa



(a) A (b) B (c) C

Figure 3.5: aaa

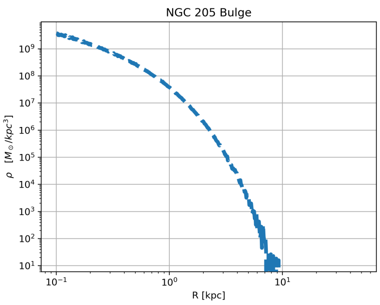
and

$$U = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j\neq i}\frac{Gm_i m_j}{|\mathbf{x_i} - \mathbf{x_j}|} \tag{3.26}$$

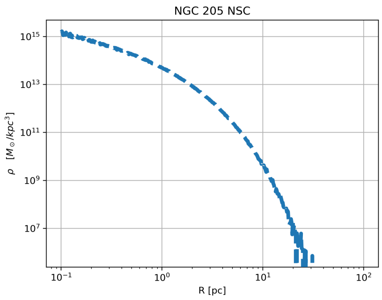
stand for the total kinetic and gravitational potential energies respsectively.
By erasing the offset due the COM kinetic energy contribute, i.e.

$$Q = \frac{2(K - K_{CM})}{|U|} \tag{3.27}$$

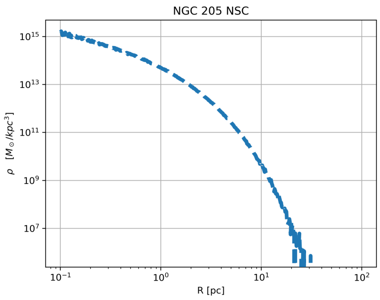
it is found $Q = x$.



(a) A

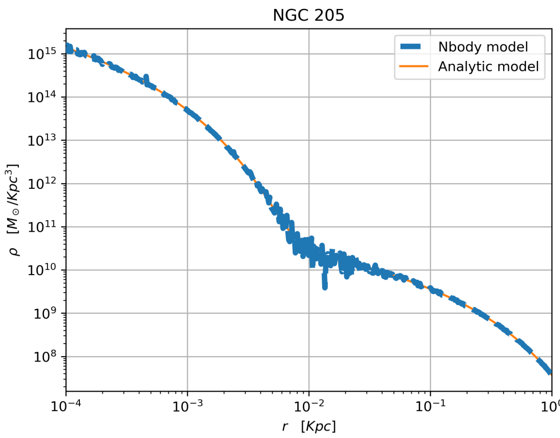


(b) B

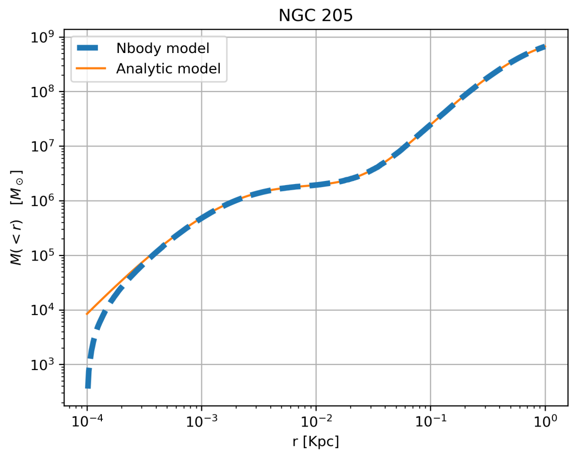


(c) C

Figure 3.6: aaa



(a) A



(b) B

Figure 3.7: aaa

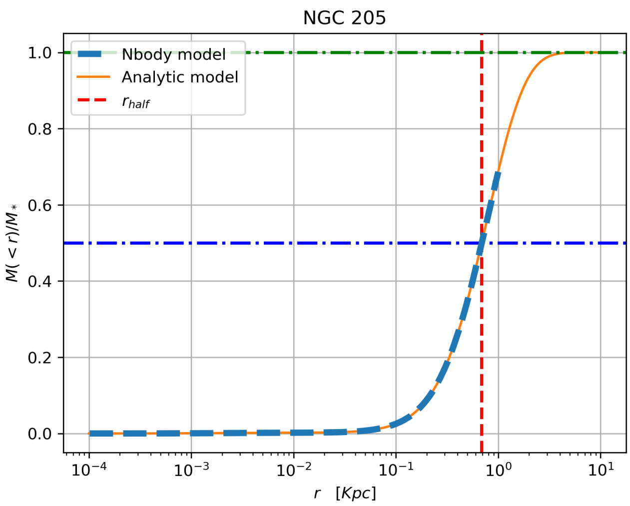


Figure 3.8: aaa

A

Physics Equation

B

Numerical Methods

2.1 Runge-Kutta (4)

Given the first order autonomous ODE

$$\frac{du}{dt} = f(u), \tag{B.1}$$

the stability anylisis of RK (4) applied to this equation leads to the following stability criterion on the time step size when $f(u) = au$:

$$\Delta t \leq \frac{2.79}{|a|}. \tag{B.2}$$

A system of first order autonomous ODEs can be written in vectorial form as:

$$\mathbf{u}' = \mathbf{f}(\mathbf{u}). \tag{B.3}$$

At linear order, the stability analysis we are allowed to perform the stability analysis to the linearized system

$$\delta\mathbf{u}' = \mathbf{J}\delta\mathbf{u}, \tag{B.4}$$

where λ_i is the i-th eigenvalue of the jacobian matrix $\mathbf{J} = \partial\mathbf{f}/\partial\mathbf{u}$.
In the eigenvectors frame, equations look like

$$\delta u_i' = \lambda_i \delta u_i \qquad \forall i = 1, N \tag{B.5}$$

and according to Equation (B.2), one would get

$$\Delta t_i \leq \frac{2.79}{|\lambda_i|} \qquad \forall i = 1, N. \tag{B.6}$$

Therefore, for a system of N Equations, a proper choice of the time step should be

$$\Delta t = \epsilon \cdot min\left\{\frac{2.79}{|\lambda_i|}\right\} \qquad \forall i = 1, N \tag{B.7}$$

with $\epsilon \leq 1$.

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